DIGITAL FORENSICS LABORATORY

Registry & Memory Analysis Unit

REGISTRY & MEMORY FORENSICS EXAMINATION REPORT

Case Identifier	AUG-2026-04-01
Report Date	04 August 2026
Examiner	Thando Sibakele Msibi
Scribe	Banel Sixole Mkhonta
Organization	Advanced School of IT
Subject System	WORKSTATION (Windows 10)
Evidence Item	RAM Dump
Classification	CONFIEDENTIAL

Executive Summary

On 04 August 2026 at approximately 10:32 UTC, a full physical memory dump and associated registry hives were acquired from a Windows 10 Pro workstation using FTK Imager. The acquisition preserved volatile evidence (RAM and pagefile) and the core registry hives (SYSTEM, SOFTWARE, SAM, and available NTUSER.DAT profiles).

Memory analysis performed with Volatility 3 (v2.3.2) identified a sophisticated Remote Access Trojan (RAT) implant active in memory. Two suspicious processes — **ratc.exe** (PID 1808) and **server.exe** (PID 3124) — exhibited anomalous parent-child relationships and were observed in a listening state on TCP (IPv4 and IPv6). The RAT remained in LISTENING mode, waiting for inbound connections, and demonstrated offset-shifting behaviour consistent with anti-forensic / stealth techniques. No legitimate Windows process was found to be the parent of these binaries.

Registry analysis (RegRipper 4.0) established that two local user accounts were present: **Thando Msibi** (RID 1002) and **Sixole Mkhonta** (RID 1003). Both accounts were created within a nine-minute window on the morning of 04 August 2026. The LogonUI key confirms Sixole Mkhonta was the actively logged-on user at the time of acquisition (LastWrite 2026-08-04 08:03:33Z). Two USB storage devices (Generic Flash Disk and SanDisk Cruzer Blade) were connected during the same short activity window.

No malicious persistence mechanisms (Run/RunOnce keys, redirected Startup folder, command-handler hijacking, anomalous services, or scheduled tasks) were identified in the available registry evidence. One legitimate OneDriveSetup Run key entry was present. Critical user-specific artefacts (UserAssist, RecentDocs, Shellbags) could not be examined because NTUSER.DAT files for the interactive users were not captured during acquisition — a significant evidence gap.

Key findings: Active RAT implant in memory with network listener; two newly created local accounts; concurrent USB activity; incomplete user-profile registry capture limiting programme-execution timeline reconstruction.

Scope of Examination

This report documents the forensic analysis of Windows registry hives and a memory image provided as part of a practical scenario. The goal is to extract artefacts that answer the following investigative questions:

- What user(s) were active on the system?
- What programs were executed?
- Is there evidence of persistence mechanisms (Run keys, services, scheduled tasks)?

- What processes existed in memory, what network connections were present, and are there indicators of code injection?

The analysis was performed using standard open-source forensic tools in a controlled laboratory environment.

Evidence Details

Registry Hives

Hive Name	Typical Path / Source	Purpose in Analysis
SYSTEM	%SystemRoot%\System32\config\SYSTEM	Services, ControlSet, USBSTOR, TimeZone, ComputerName
SOFTWARE	%SystemRoot%\System32\config\SOFTWARE	Installed programs, OS version, App Paths, NetworkList
NTUSER.DAT	C:\Users\<user>\NTUSER.DAT	User-specific Run keys, RecentDocs, UserAssist, TypedURLs
SAM	%SystemRoot%\System32\config\SAM	Local user accounts and last logon information

Memory Image

Image Filename	Memdump.mem
Image Size	
Acquisition Tool	FTK imager
OS Profile Identified	Windows 10 pro
SHA-256 Hash	

Memory Image 2

Image Filename	Win10_ram.raw
Image Size	
Acquisition Tool	FTK Imager
OS Profile Identified	Windows 10
SHA-256 Hash	

Methodology & Tools

Analysis was conducted in a controlled environment. The following tools and approaches were used:

TOOL	Version
RegRipper	4.0 main
Volatility 3	2.3.2
Volatility 2	
FTK Imager	

Analysis Approach

1. Identify the Windows version / profile from the memory image.
2. Extract and examine relevant registry keys for user activity, program execution, and persistence.
3. List running and terminated processes, examine process trees and parent-child relationships.
4. Extract network connections and sockets from memory.
5. Search for indicators of code injection / process hollowing (malfind).
6. Correlate registry and memory findings into a coherent picture of system activity.

ACQUISITION PHASE

Registry Hive Capture

In the victim's machine at 10:45:10 Mr Thando Sibakele Msibi used FTK Imager for making a memory image to analyze the registry hive set, since the machine was running, a live acquisition was performed in the machine

Figure 1.1

Launched FTK imager as administrator which a mandatory part for this process

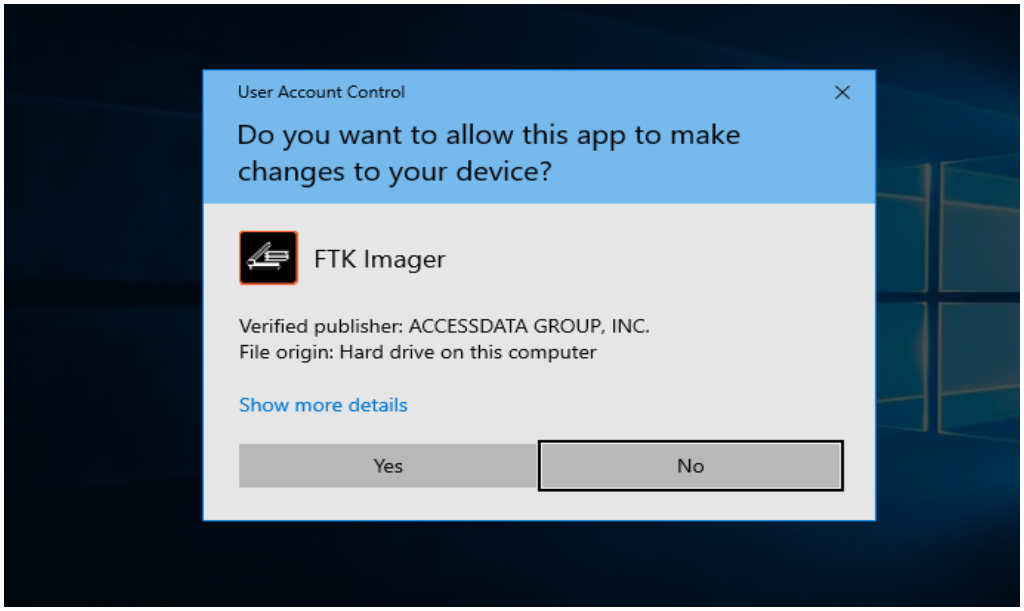
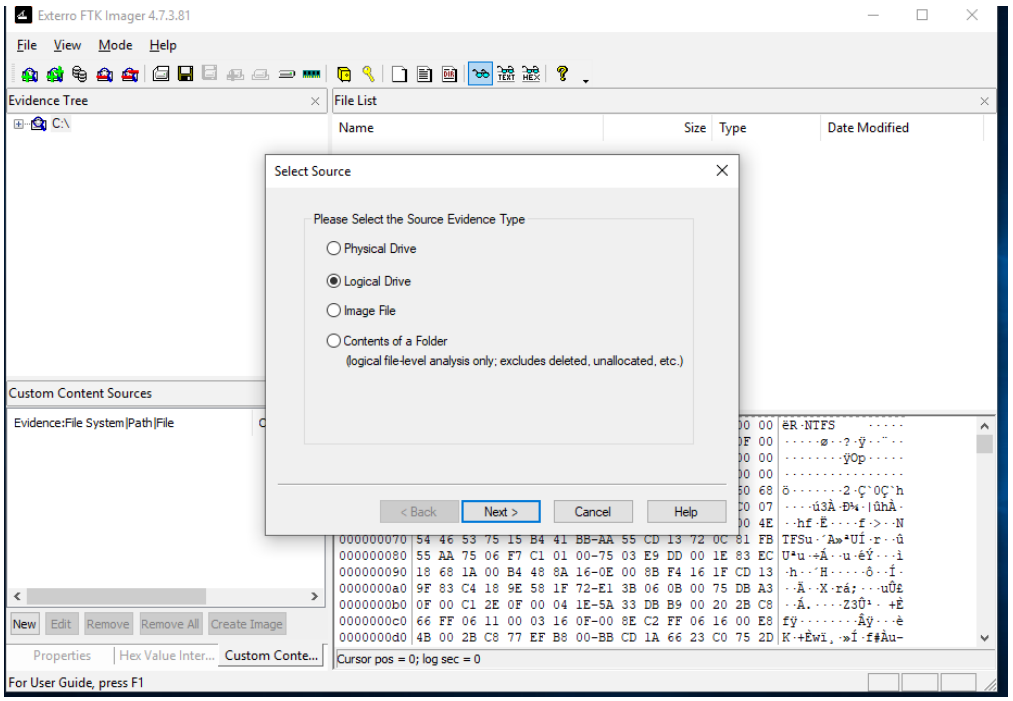


Figure 1.2



Choose Logical Drive since, imaging the entire drive was not necessary, therefore which was needed for this imaging phase was the primary partition which the C partition which contains all the registry files

Figure 1.3

Selected the obtaining protected files option and to the destination folder which the image will saved at after creation.

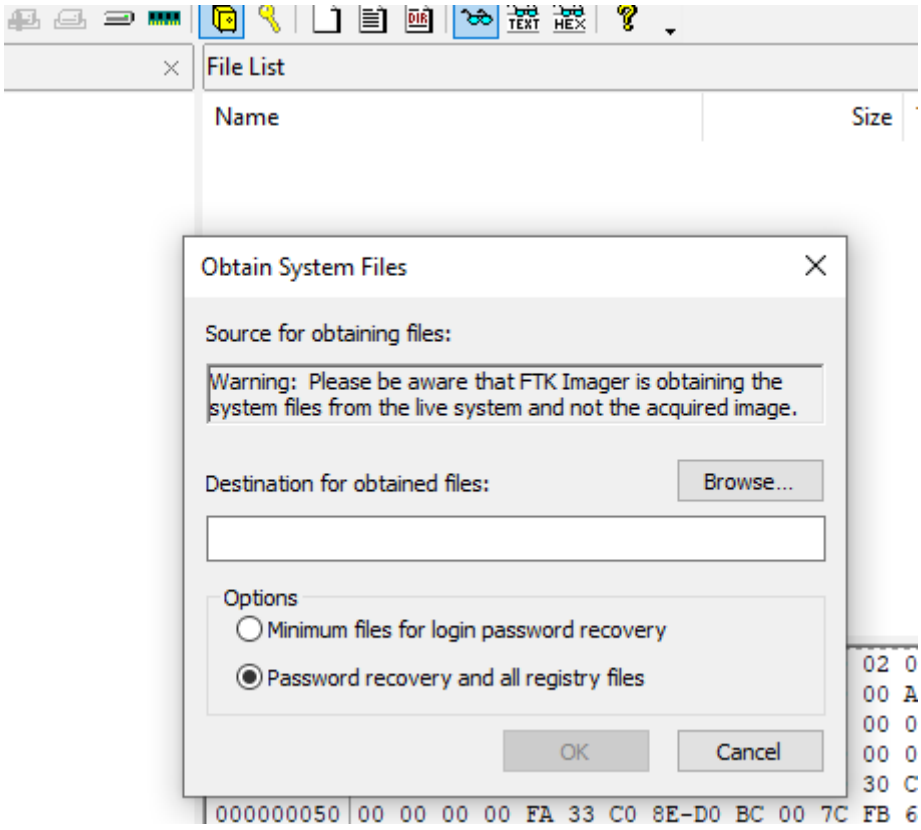


Figure 1.4

Progress of the file capture

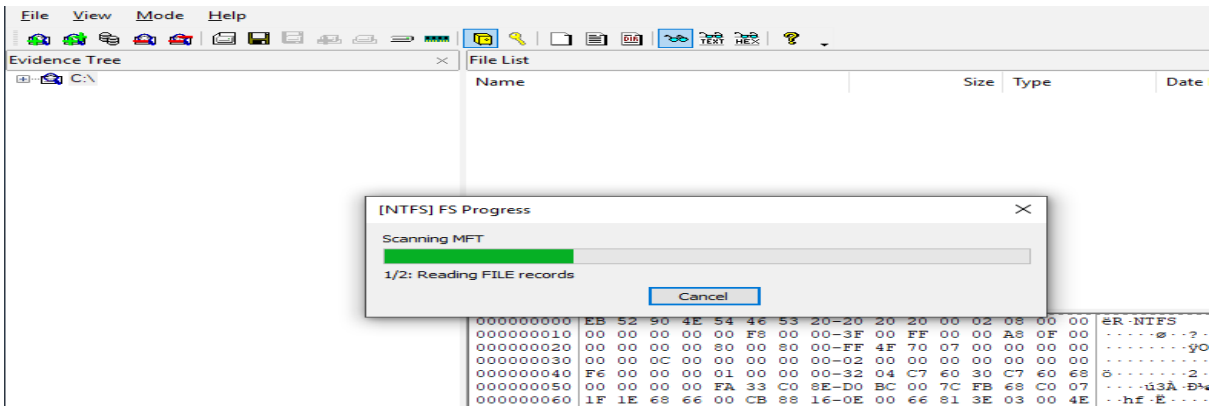


Figure 1.5

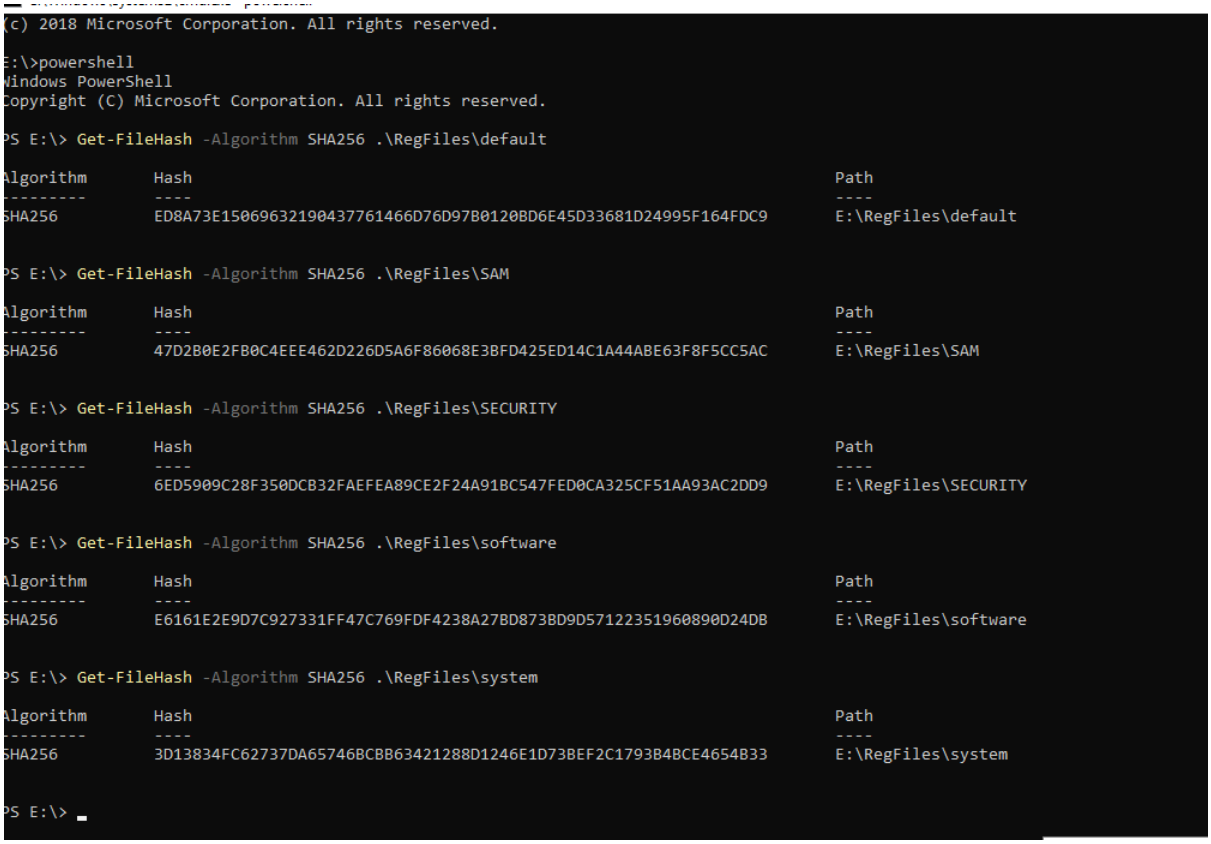
Preservation of integrity, all the files were hashed using SHA-256 algorithm and the files were saved in an external drive



Users	2026/08/04 10:58	File folder	
default	2026/08/04 17:59	File	512 KB
SAM	2026/08/04 17:59	File	64 KB
SECURITY	2026/08/04 17:59	File	64 KB
software	2026/08/04 10:03	File	67 584 KB
system	2026/08/04 17:59	File	11 520 KB

Figures 1.6

The hashed files with the SHA256 Algorithm for validation of evidence and it's integrity



```
(c) 2018 Microsoft Corporation. All rights reserved.

E:\>powershell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS E:\> Get-FileHash -Algorithm SHA256 .\RegFiles\default

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA256          ED8A73E15069632190437761466D76D97B0120BD6E45D33681D24995F164FDC9       E:\RegFiles\default


PS E:\> Get-FileHash -Algorithm SHA256 .\RegFiles\SAM

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA256          47D2B0E2FB0C4EEE462D226D5A6F86068E3BFD425ED14C1A44ABE63F8F5CC5AC       E:\RegFiles\SAM


PS E:\> Get-FileHash -Algorithm SHA256 .\RegFiles\SECURITY

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA256          6ED5909C28F350DCB32FAEFEA89CE2F24A91BC547FED0CA325CF51AA93AC2DD9       E:\RegFiles\SECURITY


PS E:\> Get-FileHash -Algorithm SHA256 .\RegFiles\software

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA256          E6161E2E9D7C927331FF47C769FDF4238A27BD873BD9D57122351960890D24DB       E:\RegFiles\software


PS E:\> Get-FileHash -Algorithm SHA256 .\RegFiles\system

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA256          3D13834FC62737DA65746BCBB63421288D1246E1D73BEF2C1793B4BCE4654B33       E:\RegFiles\system


PS E:\>
```

Memory Image Capture

In memory capturing, FTK imager played a vital role also since in order to prevent alteration of evidence in the suspect's machine FTK was used via usb as a plug and play element to make a memory image in the suspect machine

Figure 2.1

Path to the FTK Imager folder, where the whole folder was copied since it was already extracted and that will give the edge of making the usb drive the play and play role which was required

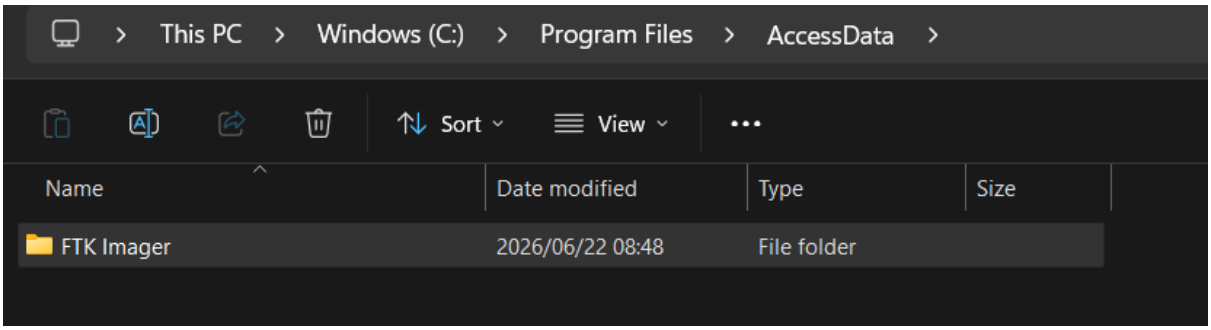


Figure 2.2

Copying the FTK folder to the usb drive

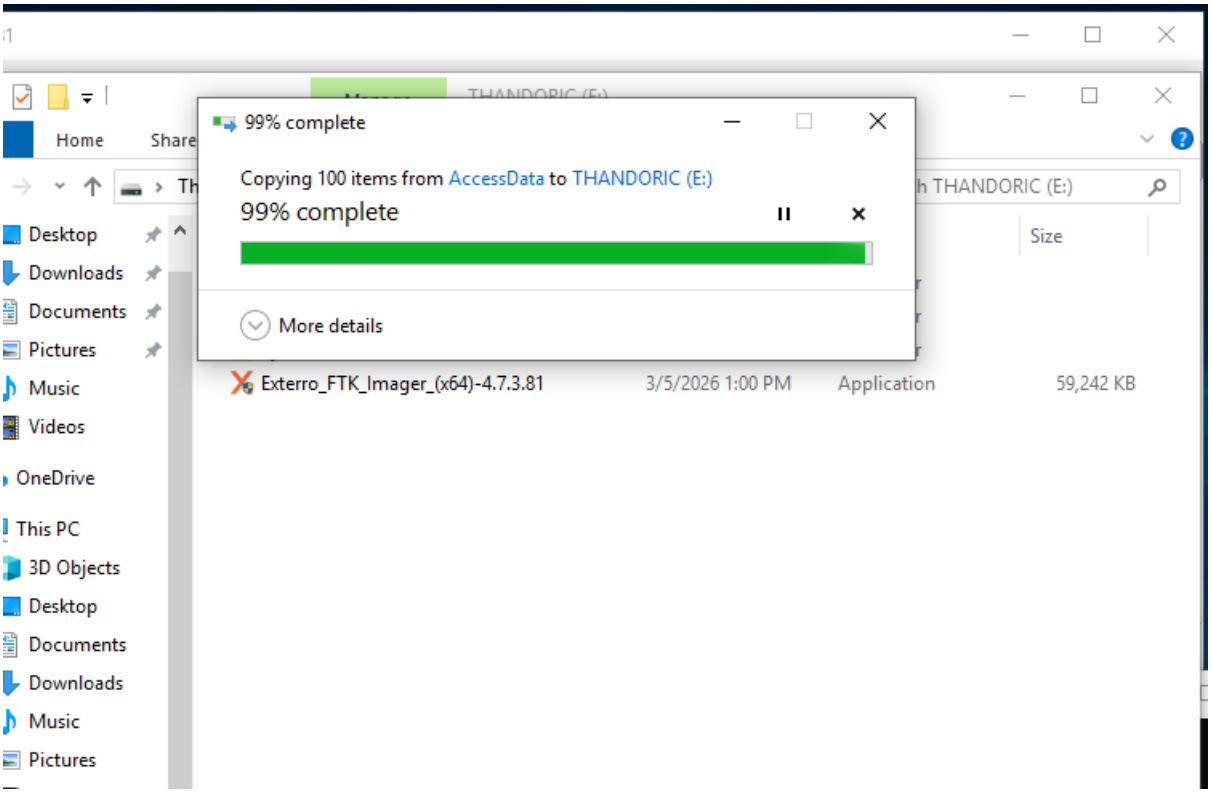


Figure 2.3

After copying the FTK folder in the usb drive

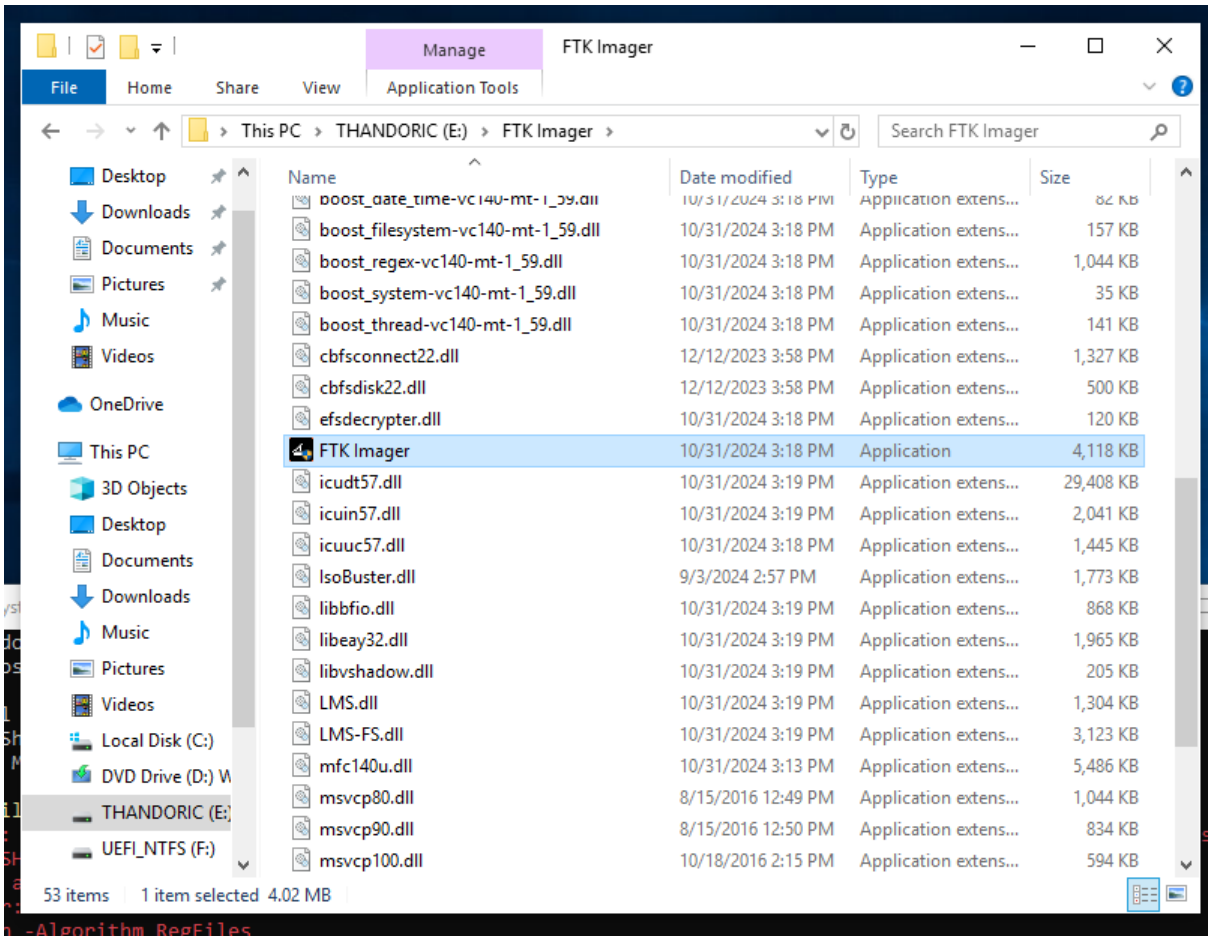


Figure 2.4

As done in the registry hive capturing, always run FTK imager as an administrator since, when the tool works in the administrative privilege it draws out its full power with having limitation compared being used as user.

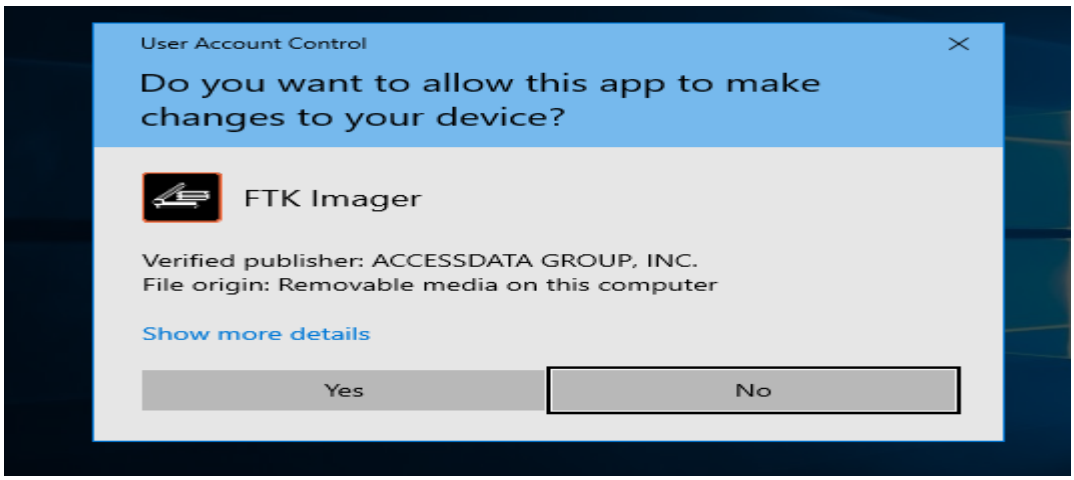


Figure 2.5

Inside FTK imager clicked capture memory since it a memory image creation, it uses the live acquisition method and memory is volatile as the golden rule states start with volatile evidence such as memory since once turned off data is lost for good.

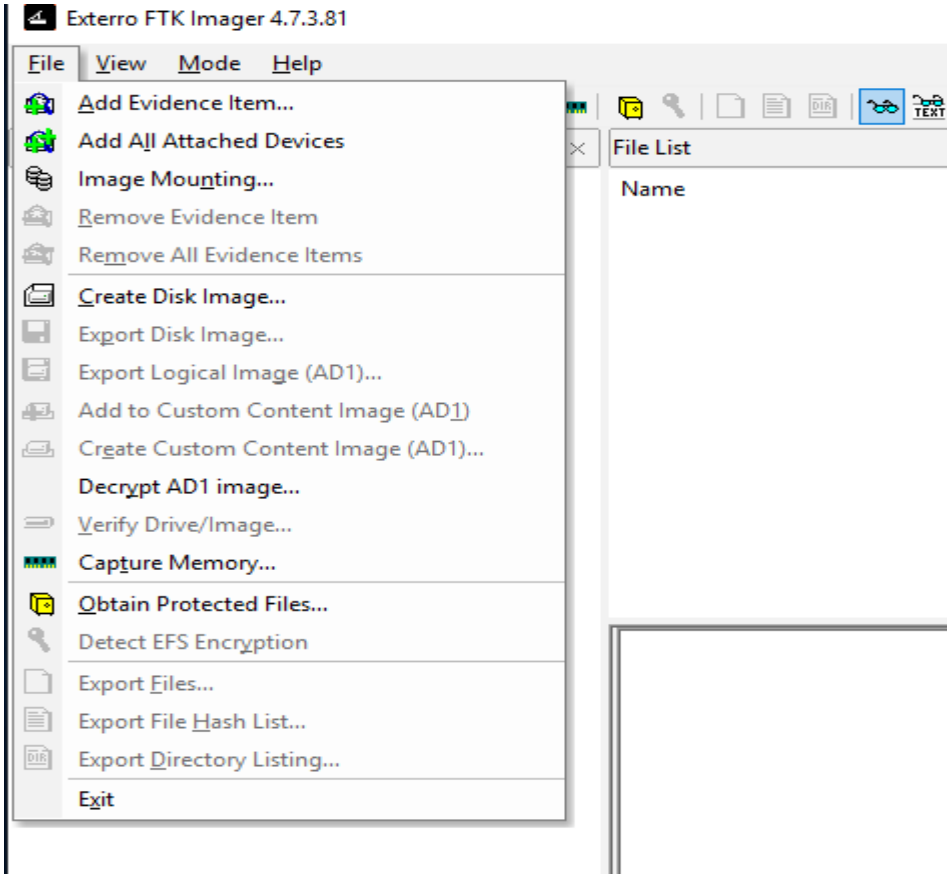


Figure 2.6

Included the pagefile and create an AD 1 file since pagefile.sys is the windows' swap file, when RAM is full, windows move data to disk temporarily. Which often contains fragments of process, strings or even passwords that were swapped out of the RAM.

AD1 is AccessData's Logical Image Format. It's a container file that stores dump plus metadata. Preserves evidence integrity with hashes and keeps everything in one structed file instead of loose. mem or. raw dumps

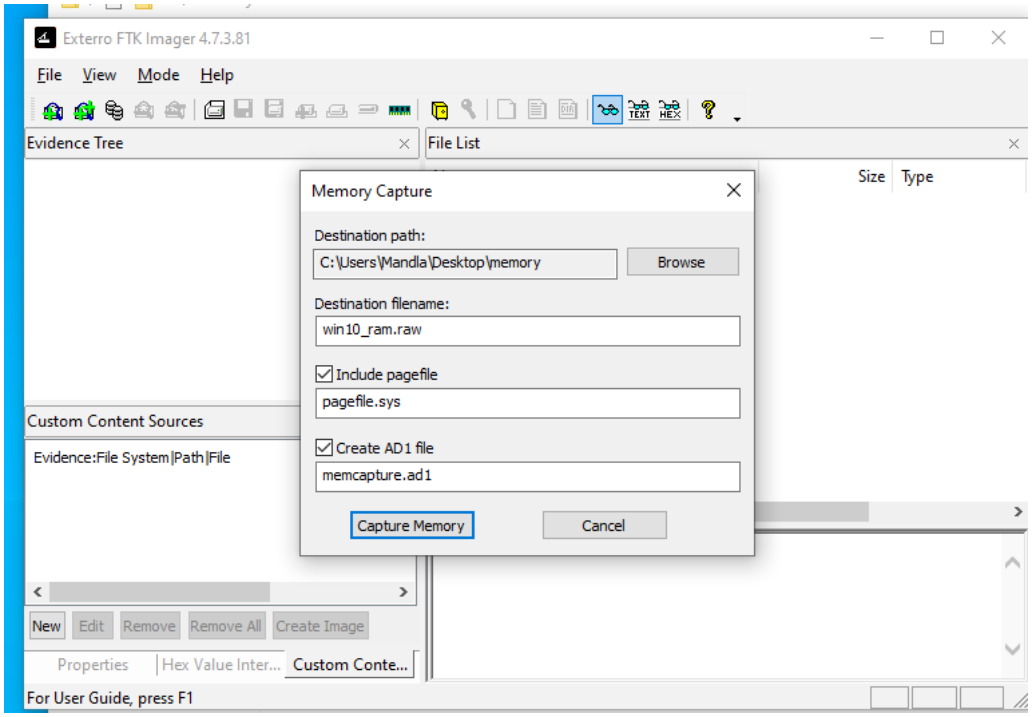
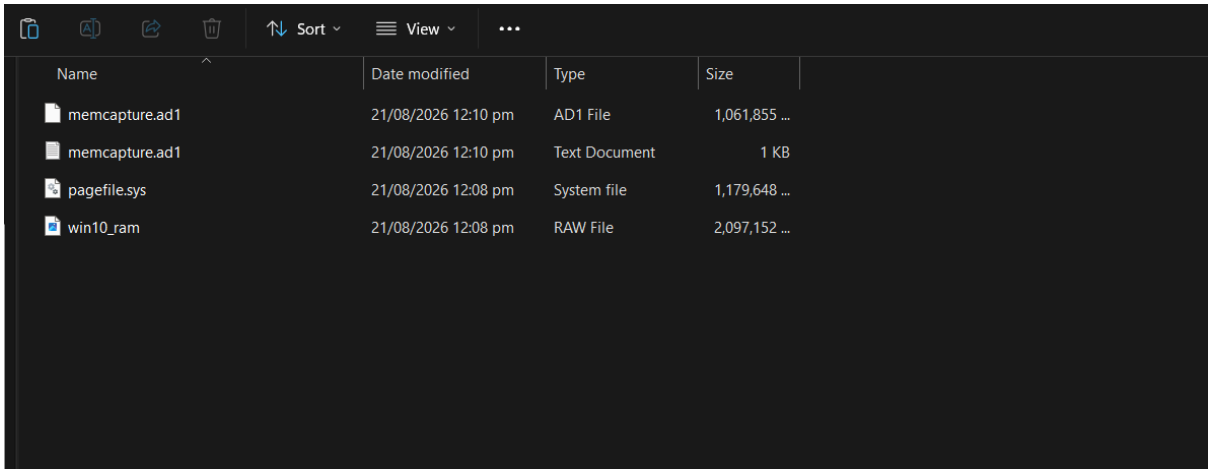


Figure 2.7

After completion the MemoryDump or image, the folder contained the pagefile.sys, memcapture.ad1and the win10_ram.raw and also the memcapture.ad1.txt which contains information about the dump file when it started and ended the exact time. It contains the integrity checks too



Analysis - 1

Registry Forensics

At 12:30:22 Mr Thando Sibake Msibi started the registry hive set analysis. Registry is a system database used a windows operating system to store system configuration and user activities. It is one of the juiciest sources of evidence for digital investigators.

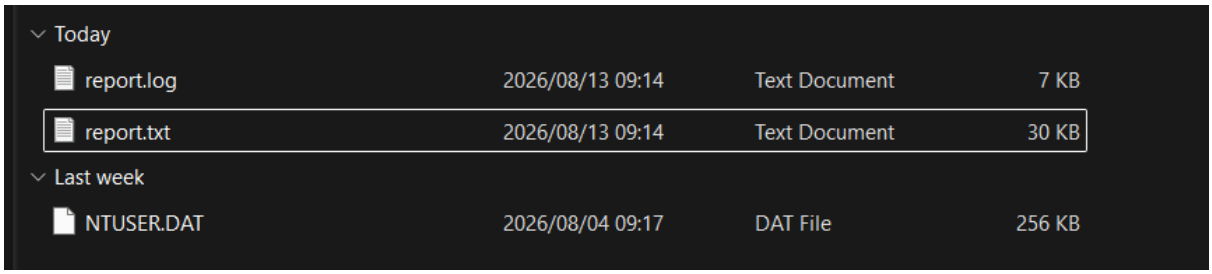
Registry are in hieracia way. First is the hives followed by key going fourth to subkeys which are keys under keys. This analysis is done for the victim's machine

Firstly, navigated inside the folder and opened cmd then typed the command

"reg.exe -r NTUSER.DAT -p ntuser > report.txt" which was piping the results to report.txt.

```
C:\Users\Thando Sibakhele\Downloads\RegRipper4.0-main\RegRipper4.0-main>rip.
exe -r "C:\Users\Thando Sibakhele\Downloads\RegRipper4.0-main\RegRipper4.0-m
ain\RegFiles\Users\Default\NTUSER.DAT" -f ntuser > ntuser_report.txt
```

Figure 3.1



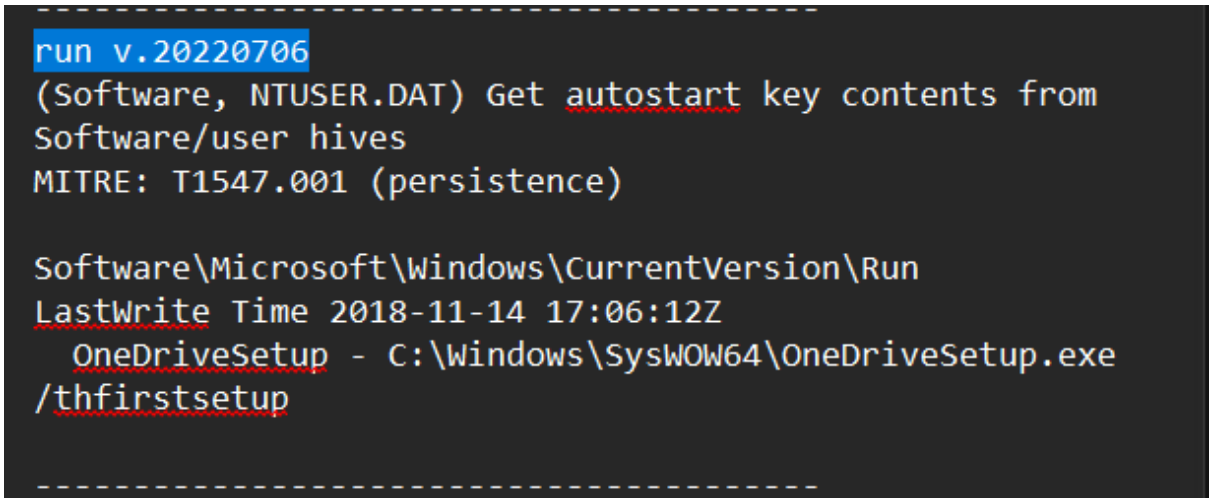
Report.txt created and opened the report.txt searched for "run".

Found –

Key: Software\Microsoft\CurrentVersion\Run [cite: 1] Artifact / Entry Name: OneDriveSetup [cite: 1] Executable Path:

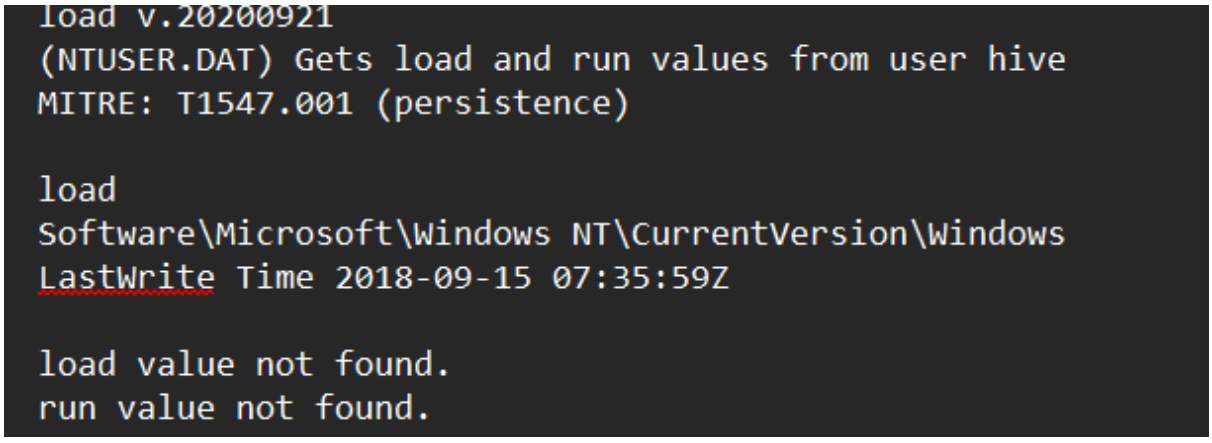
C:/Windows\SysWOW64\OneDriveSetu.exe /thfirstsetup [cite: 1] Last Modified Timestamp: ** '2018-11-14 17:17:06:12Z' [cite: 1] * **MITRE ATT&CK Technique: **T1547.001 (Boot or Logon AutoStart Execution: Registry Run Keys / Startup Folder) [cite: 1]

Figure 3.2



Only OneDrive setup was set to AutoStart.

Figure 3.3



Confirmation that no secondary legacy ‘load’ or ‘run’ execution binaries were set under the ‘Windows NT\CurrentVersion\Windows’ subkeys [cite: 1].

The default Startup folder remains untampered with and points to the standard location (‘%USERPROFILE%\AppData\Roaming\Microsoft\Windows\StartMenu\Programs\Startup’) [cite: 1].

Figure 3.5

Output of the RegRipper’s “shellfolders” plugin run against the SOFTWARE hive.it resolves the actual filesystem path bound to the User Shell Folder “Starup” value – the value – the location Windows checks every logon for programs to auto-launch. The results confirm the StarUp folder still points to the defaults Windows location (%USERPROFILE%\AppData\Roaming\Windows\Start Menu\Programs\Startup) and has not been redirected to an attacker-controlled folder, a technique

some malware uses to hides autorun items outside the normal, well-monitored path. No tampering was found here.

Since this is the 'Default' user profile 'NTUSER.DAT' ('...\RegFiles\Users\Default\NTUSER.DAT'), persistence mechanisms installed for specific active user accounts will be found in those specific user' 'NTUSER.DAT' files, as well as the machine-wide 'SYSTEM' and 'SOFTWARE' registry hives [cite: 1]

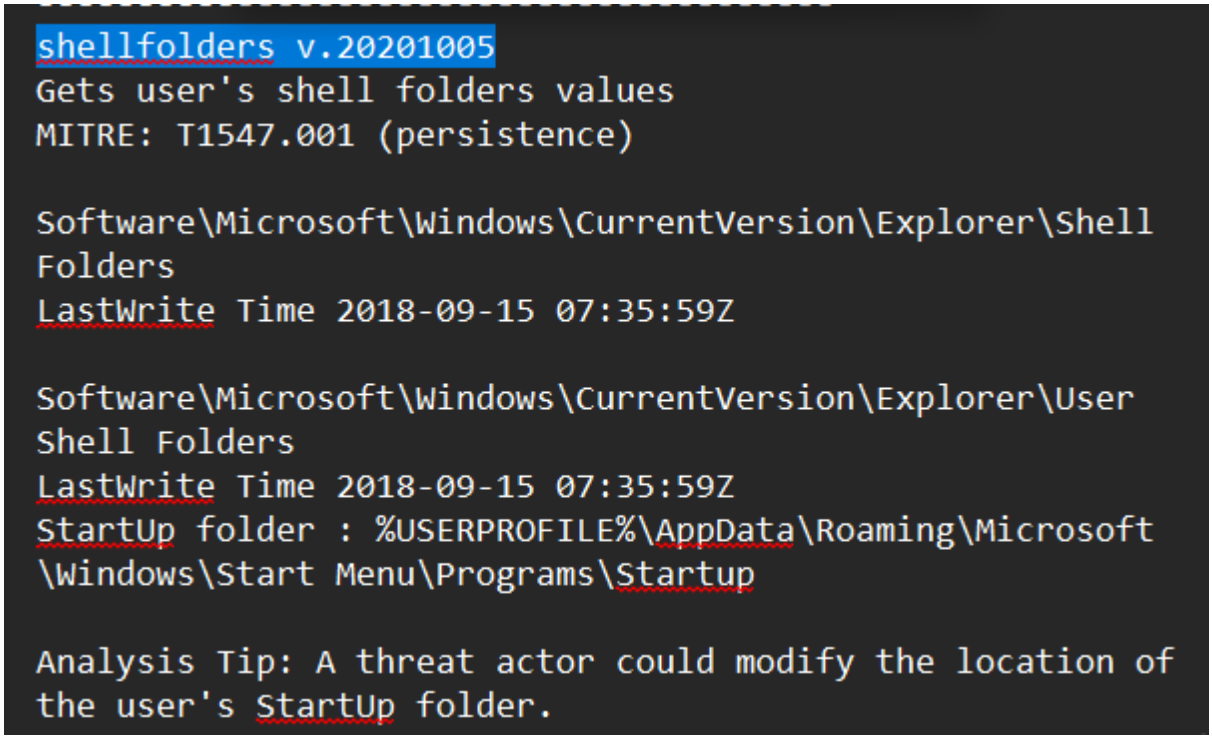
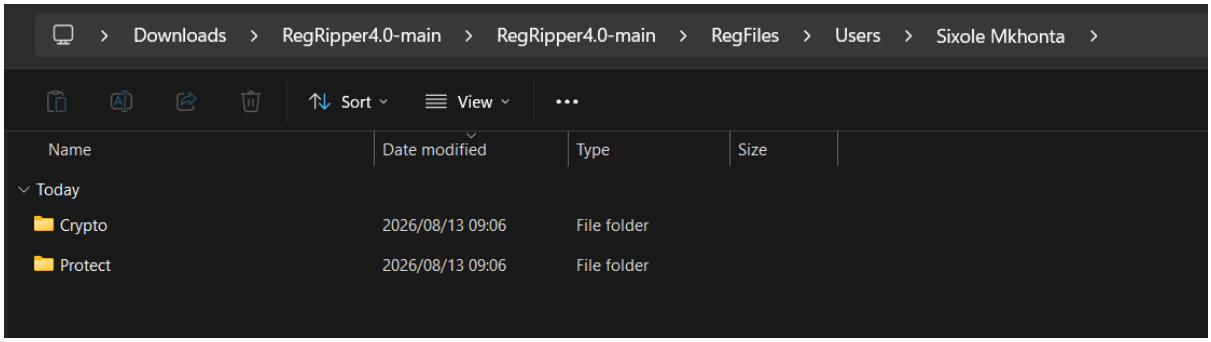
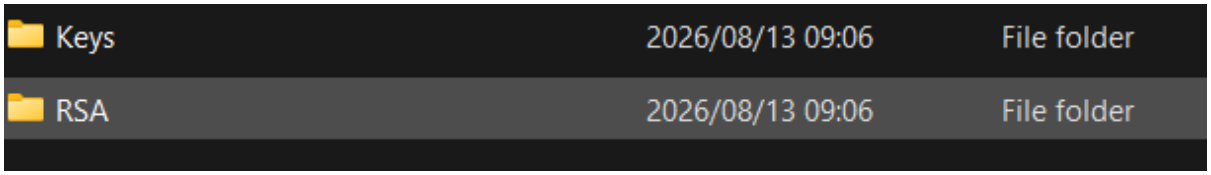


Figure 3.6



No NTUSER.DAT files were created for user Thando during the capture of the reg hives. Browsing to RegFiles\Users\Thando Msibi shows this profile folder is empty – no NTUSER.DAT was capture for this account during acquisition. This means Run/RunOnce keys, RecentDocs, and other user-specific artefacts belonging to Thando Msibi cannot be examined from this evidence set; only machine-wide SAM, SYSTEM and SOFTWARE hives, and the Defaults profile hive, are available for this account.

Figure 3.7



Also, no NTUSER.DAT files for user Sixole, by contrast the profile folder for Sixole Mkhonta is not empty – it contains two sub-folders named “Crypto” and “Protect”, both dated 2026/08/13. These folder names stand-out and are worth flagging for follow-up, since they could correspond to a password manager, encryption utility or cryptocurrency wallet application data carried over with the profile, or they may simply hold the NTUSER.DAT and related hive files for this account.

We opened both folders and capture a directory listing of their contents, and confirms that an NTUSER.DAT files is present inside either one, results only show the keys and RSA folders, since the name alone suggest encryption or crypto-wallet activity of which they deserve closer examination

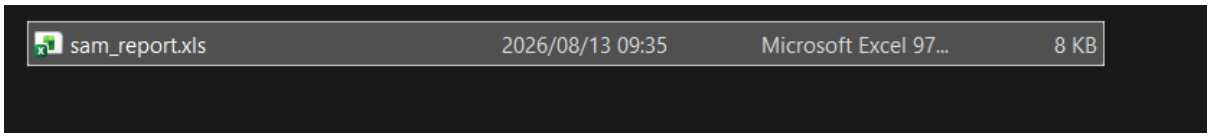
Figure 3.8

Figure 3.9

```
C:\Users\Thando Sibakhele\Downloads\RegRipper4.0-main\RegRipper4.0-main>rip.
exe -r "C:\Users\Thando Sibakhele\Downloads\RegRipper4.0-main\RegRipper4.0-m
ain\RegFiles\SAM" -p samparse > sam_report.xls
Launching samparse v.20200825
```

RegRipper’s command-line tool (rip.exe) is run against the SAM hive using the “samparse” plugin, with output piped to sam_report.xls. The SAM (Security Account Manager) hive stores local user account information – usernames, relative identifies (RIDs), account creation and last-login timestamps, and account flags – making it the primary source for answering “what user(s) were active” on this machine.

Figure 4.0



The command completed successfully and produced sam_report.xls (8 KB) in the working directory, ready to be opened for review.

Figure 4.1

76	Username : Thando Msibi [1002]
77	Full Name :
78	User Comment :
79	Account Type :
80	Account Created : 2026-08-04 07:14:01Z
81	Name :
82	Last Login Date : 2026-08-04 07:14:03Z
83	Pwd Reset Date : 2026-08-04 07:14:01Z
84	Pwd Fail Date : Never
85	Login Count : 1
86	Embedded RID : 1002
87	--> Password does not expire
88	--> Password not required
89	--> Normal user account
90	
91	ForcePasswordReset : 0
92	
93	Username : Sixole Mkhonta [1003]
94	Full Name :
95	User Comment :
96	Account Type :
97	Account Created : 2026-08-04 07:23:43Z
98	Name :
99	Last Login Date : 2026-08-04 08:03:33Z
100	Pwd Reset Date : 2026-08-04 07:23:43Z
101	Pwd Fail Date : Never

Opened sam_report.xls reveals two local user accounts on this machine, which directly answers the “what users(s) were active “question:

Thando Msibi (RID 1002) -account created 2026-08-04 07:14:01Z, last login 2026-08-04 07:14:03Z, login count 1. Flags: password does not expire, password not required, normal user account

Sixole Mkhonta (RID 1003) – account created 2026-08-04 07:23:43Z, last login 2026-08-04 08:03:33Z.

Both accounts were created within about nine minutes of each other on the same morning and the “password not required” / “password does not expire” flags on Thando Msibi’s account are worth nothing in the report was a weak security configuration. Sixole Mkhonta’s later login timestamp (08:03:33Z, versus 07:14:03Z for Thando Msibi) suggest this was independently by the logon UI key evidence

Figure 4.2

To get broader coverage than a single plugin, the RegRipper GUI (a graphical front end for rip.exe) was pointed at the SOFTWARE hive and run against the full plugin catalogue (appaths, appsetup, assoc, audiodev, auth, autoadminlogon, autorun, blm, btconfig, calibrator, certpadding, certs, clipbrd, and many more, shown scrolling in the output pane). Running every plugin in one pass, rather than one plugin at a time, ensures no relevant autostart, application, or configuration artefact is missed - including the "autorun" plugin, which specifically enumerates Run/RunOnce-style persistence locations across the hive.

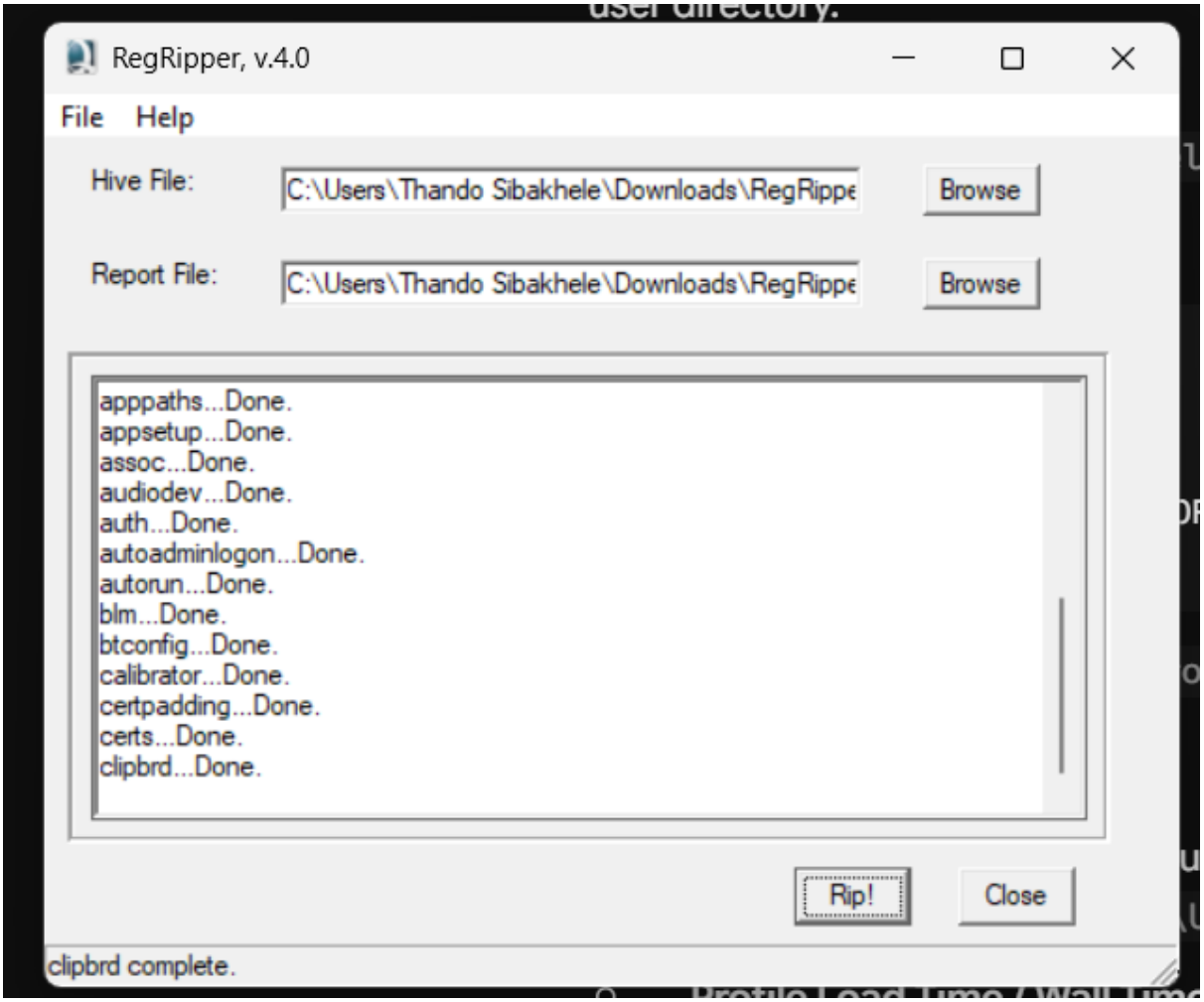
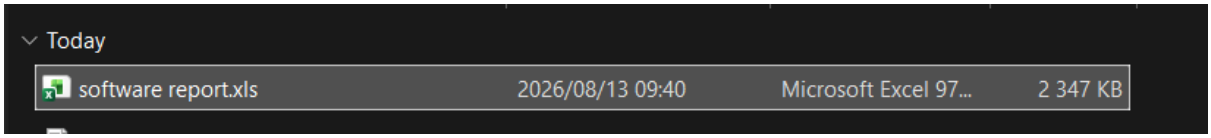
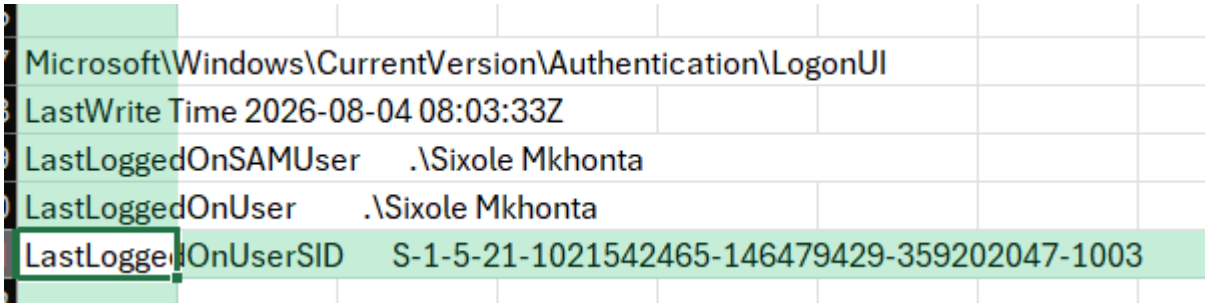


Figure 4.3



This full-plugin run against SOFTWARE produced software report.xls (2,347 KB) - substantially larger than the single-plugin SAM report, since it contains the combined output of every plugin that matched a key present in the hive.

Figure 4.4



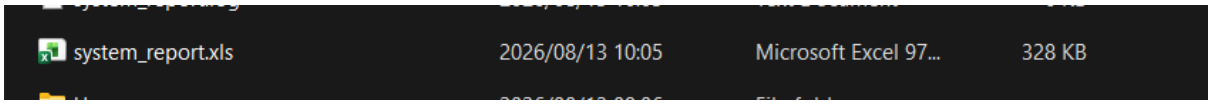
The Logon UI key (Software\Microsoft\Windows\CurrentVersion\Authentication\LogonUI) records which account was last displayed and selected on the Windows logon screen. Here it confirms Sixole Mkhonta (SID S-1-5-21-1021542465-146479429-359202047-1003) as both the LastLoggedOnSAMUser and LastLoggedOnUser at the time of capture, with a LastWrite time of 2026-08-04 08:03:33Z. This directly answers "what user was active": Sixole Mkhonta was the account actively logged on when the registry hives and memory image were captured.

Figure 4.5



The "cmd_shell" plugin checks the shell\open\command values for common file-type and protocol handlers - here, Internet Explorer's registered handler - for evidence of command hijacking, a technique where malware appends extra commands to a legitimate handler so its own code runs whenever that handler is invoked. The Cmd value shown is the expected, unmodified path (C:\Program Files\Internet Explorer\iexplore.exe) with nothing appended, so no hijacking was found. The "Error: Undefined subroutine &main: alertMsg" line underneath is an unrelated bug in this build of the plugin and does not affect the validity of the Cmd value reported above it.

Figure 4.6



The SYSTEM hive was processed next, again using the full RegRipper plugin set, producing system_report.xls (328 KB). The SYSTEM hive holds machine-wide configuration data, including installed services, device and USB connection history, network interfaces, and the computer name and time zone - the usbstor plugin output reviewed next is drawn from this report.

Figure 4.7

The "usbstor" plugin parses HKLM\SYSTEM\CurrentControlSet\Enum\USBSTOR and lists every removable storage device ever connected to this machine. Two devices were found:

- A generic "Flash Disk" (Generic Flash Disk USB Device) - first installed/inserted 2026-08-04 07:28:07Z, last inserted 2026-08-04 08:02:55Z.
- A SanDisk Cruzer Blade (serial 4C530001200320100273&0) - first installed/inserted 2026-08-04 07:45:39Z, last inserted 2026-08-04 08:03:01Z.

Both devices were connected within the same short window as the two user account creations and the LogonUI write time seen earlier (roughly 07:14Z-08:03Z), suggesting the USB activity, the account creation, and the observed logon session are all part of the same short window of activity on this machine and we will use that for building an overall timeline of events.

```
usbstor v.20220524
(System) Parses Enum\USBStor key

Disk&Ven_Generic&Prod_Flash_Disk&Rev_8.07
 0FD4CB7CE6E7B4D49E43&0
  DeviceDesc    : @disk.inf,%disk_devdesc%;Disk drive
  Mfg       : @disk.inf,%genmanufacturer%;(Standard disk drives)
  Service     : disk
  FriendlyName  : Generic Flash Disk USB Device
  First Install  : 2026-08-04 07:28:07Z
  First Inserted : 2026-08-04 07:28:07Z
  Last Inserted  : 2026-08-04 08:02:55Z

Disk&Ven_SanDisk&Prod_Cruzer_Blade&Rev_1.00
 4C530001200320100273&0
  DeviceDesc    : @disk.inf,%disk_devdesc%;Disk drive
  Mfg       : @disk.inf,%genmanufacturer%;(Standard disk drives)
  Service     : disk
  FriendlyName  : SanDisk Cruzer Blade USB Device
  First Install  : 2026-08-04 07:45:39Z
  First Inserted : 2026-08-04 07:45:39Z
  Last Inserted  : 2026-08-04 08:03:01Z
```

Figure 4.8

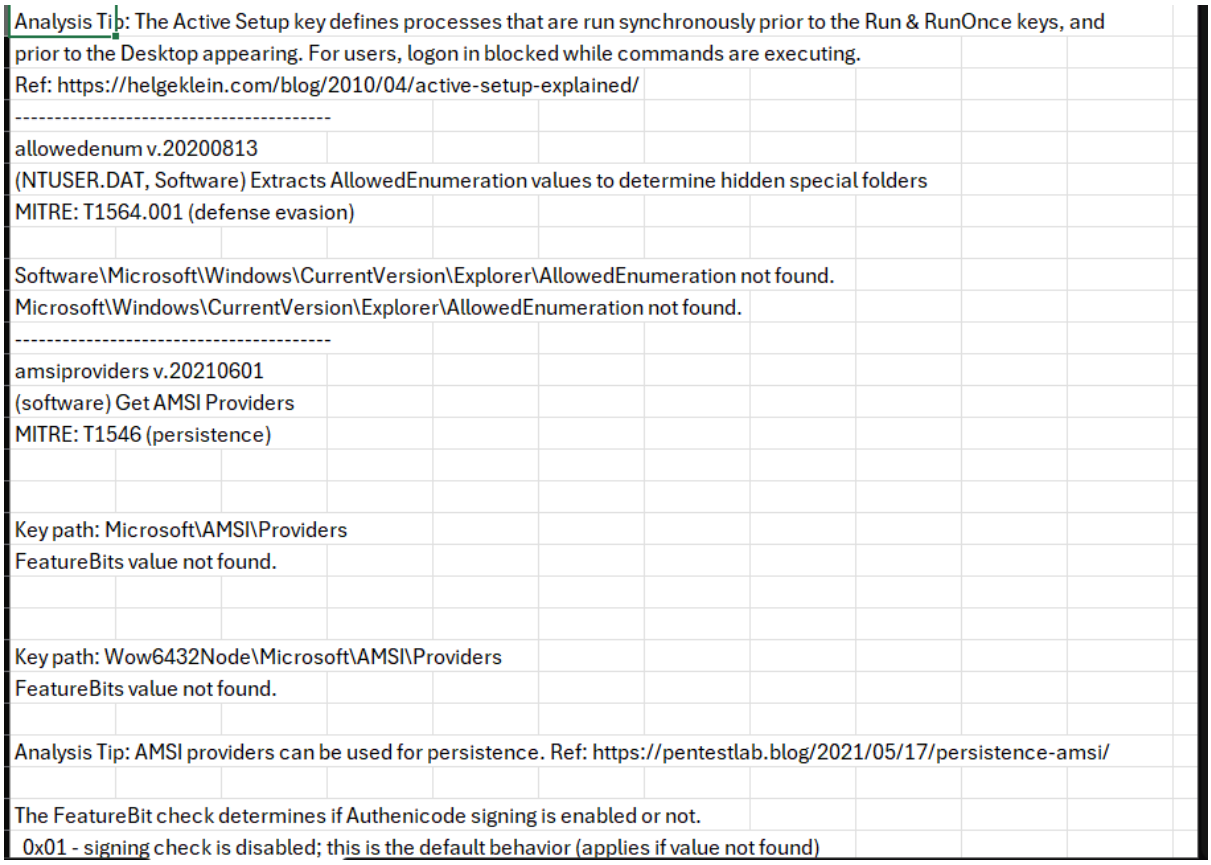
```
Analysis Tip: Look for MSFT Virtual_DVD-ROM devices that map to drive letters, as well as USB devices.

Volumes
\DosDevices\F:  Volume GUID: {6aac517d-883d-4f98-849e-27b03e7bde2a}
\??\Volume{83ce9418-901d-11f1-905f-dc4ba1682e3b} Volume GUID: {b1bbc6b5-cb5c-430f-8da5-528b32691cf9}
\??\Volume{83ce9571-901d-11f1-905f-dc4ba1682e3b} Volume GUID: {1493d753-bd3e-4369-a6c5-35e8dd230fa1}
\DosDevices\G:  Volume GUID: {9828f1f1-ad0d-499a-b1e2-34639dfa4552}
\DosDevices\E:  Volume GUID: {b1bbc6b5-cb5c-430f-8da5-528b32691cf9}
\DosDevices\H:  Volume GUID: {1493d753-bd3e-4369-a6c5-35e8dd230fa1}
\??\Volume{83ce9570-901d-11f1-905f-dc4ba1682e3b} Volume GUID: {9828f1f1-ad0d-499a-b1e2-34639dfa4552}
\??\Volume{83ce941a-901d-11f1-905f-dc4ba1682e3b} Volume GUID: {6aac517d-883d-4f98-849e-27b03e7bde2a}
```

This volume/mounted-devices information ties each device's Volume GUID to the drive letter Windows assigned it (for example \DosDevices\F: \DosDevices\G: \DosDevices\H:). Cross-referencing these GUIDs against the two USBSTOR entries in Figure 32 lets an investigator state, with evidence, which specific drive letter each physical USB device was mounted as this will help to spot if a suspicious

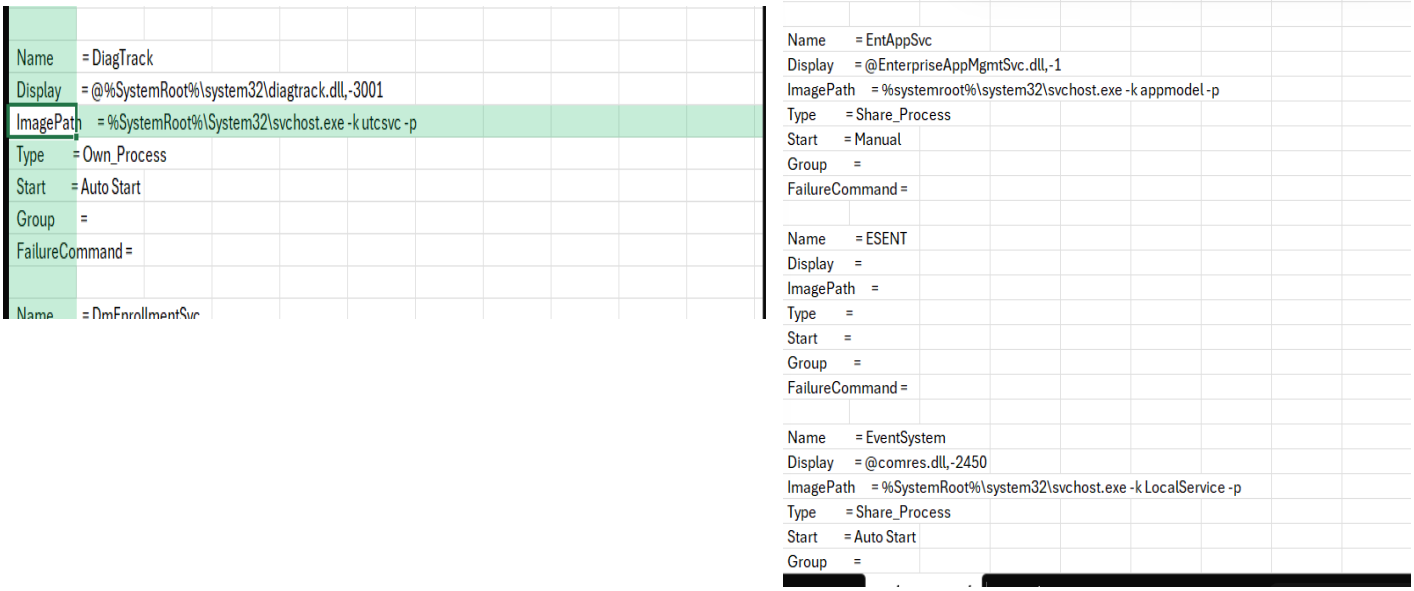
file is later found to have been accessed from, or copied to, a particular drive letter during the memory or file-system analysis

Figure 4.9



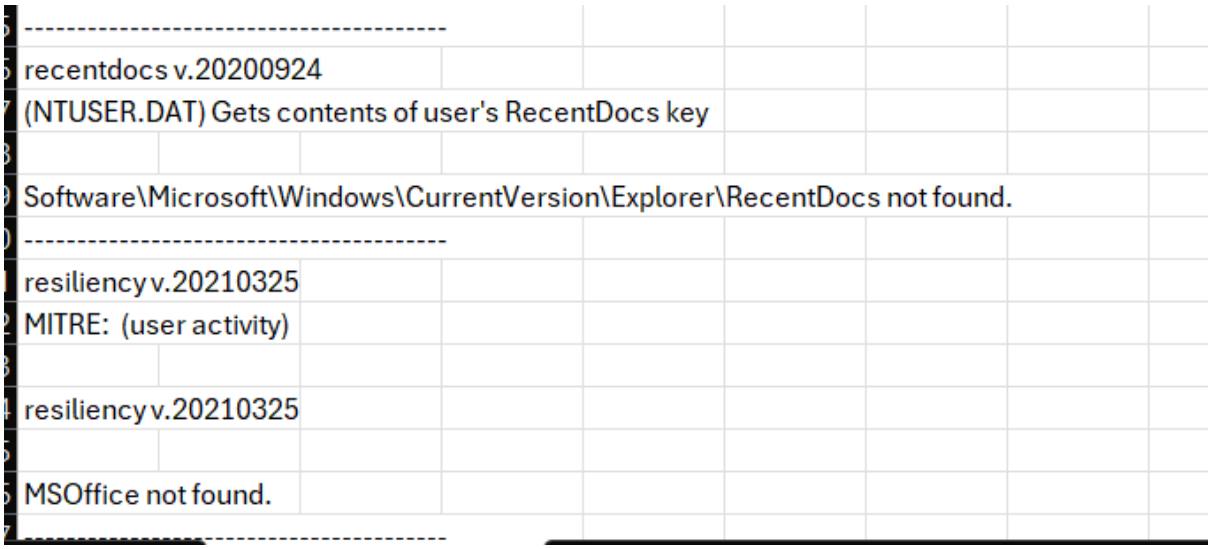
The RunOnce Key

Figure 5.0



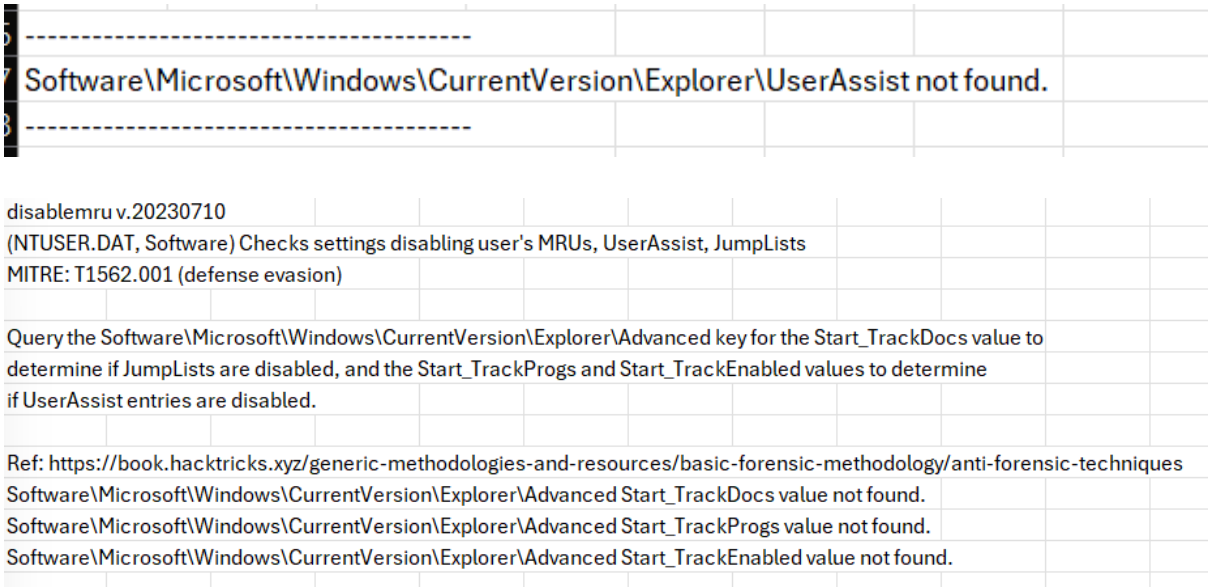
These were the services and there were no scheduled tasks

Figure 5.1



No recent docs were found

Figure 5.2 & 5.3



Shellbags were not found either. This is because NTUSER.DAT for each user is not available we were searching for the default ntuser.dat. in real work environment ntuser.dat for each user need to be captured and analyzed for shellbags, recent docs and usserassit. Shellbags are registry entries (stored under NTUSER.DAT/USRCLASS.DAT) that record the folders a user has browsed in Windows Explorer - including their view settings and full path - and they persist even after the folder itself is deleted or the drive is removed, making them useful for proving a user navigated to a specific local or removable-drive path. UserAssist is a registry key that logs every program launched from the GUI (Explorer/Start Menu), together with a run count and last-executed timestamp; its values are ROT13-encoded, which RegRipper decodes automatically, and it is normally the strongest single registry-based source of evidence for "what programs were executed." RecentDocs records the files and folders a user has most recently opened, broken down by file

extension/application, and can show what documents were accessed even if they have since been deleted.

Analysis-2

Memory Forensics

At 09:09:23 Mr Sixole Mkhonta started the memory analysis for the suspect machine. As known the golden rule in digital forensics volatile evidence is crucial and start with and it contains a lot of important information to be used as evidence. The Ram holds all memory of the machine data since everything passes through it at that particular login session of the user.

This will answer the questions of who is the user, what process they were running and the timestamps of those processes

The most important fact to avoid any aquitaries of case is to prove the aquamarines preservation of integrity of the data to analyzed

Figure 6.1

```
Information for C:\Users\Mandla\Desktop\memory\memcapture.ad1:
[Computed Hashes]
 MD5 checksum:    35246d5b5c2e2ed58fc29ecc2bad3c8a
 SHA1 checksum:   61e4982ec4b7a520ac9667abd9a81dee5059cee2
```

FTK imager verification of preservation integrity proves the data was not altered it is intact

Figure 6.2

```
┌──(mrvoidhalo㉿kali)-[/run/media/mrvoidhalo/New Volume1/memory]
└─$ sha256sum memcapture.ad1 memcapture.ad1.txt pagefile.sys win10_ram.raw
8f75ebb7d39cb4268ecd1bac5e6204f3691a0b382a13a937385a3459988029fa  memcapture.ad1
fafc5c05c7ca1e2cdcc163fbf29fbdc752f20f49ff76a0d2c97de07a83d4b390  memcapture.ad1.txt
1bf41a2f86126a4d4f3e7b223dcd5d2905063fb7dd3d9581d3d965f7de58ea38  pagefile.sys
37343f2ea86ef9fb91e5bfa997854ec48525feaaca3745563f68ebe35ff0a7d8  win10_ram.raw
```

At 09:25:09Z verified the memory dump for in Linux (Kali) and if gave different sha256 hashes since its kali-Linux cannot do a verification on all the files as a one thing but it does them individually

Figure 6.3

```
┌──(mrvoidhalo㉿kali)-[~/volatility3]
└─$ python3 vol.py -f ~/Desktop/memory/win10_ram.raw windows.info
Volatility 3 Framework 2.28.2
Progress:  100.00               PDB scanning finished
Variable        Value

Kernel Base     0xf8005b208000
DTB     0x1ad000
Symbols file:///home/mrvoidhalo/volatility3/volatility3/symbols/windows/ntkrnlmp.pdb/11BC9A513F1140CA359ECDF50F0122C1-1.json.xz
Is64Bit True
IsPAE   False
layer_name      0 WindowsIntel32e
memory_layer    1 FileLayer
KdVersionBlock  0xf8005b6303b8
Major/Minor     15.18362
MachineType     34404
KeNumberProcessors      2
SystemTime      2026-08-21 10:08:04+00:00
NtSystemRoot    C:\Windows
NtProductType   NtProductWinNt
NtMajorVersion  10
NtMinorVersion  0
PE MajorOperatingSystemVersion  10
PE MinorOperatingSystemVersion  0
PE Machine      34404
PE TimeDateStamp        Sat May 30 21:12:57 1998
```

Then at 09:30:02 Mr Sixole Mkhonta used volatility3 in kali Linux which is a powerful tool for analysing memory in memory forensics

The first command that was run was the windows.info command which gives the information about the OS running, the version of the OS

Then in the figure above it of the screenshot it shows the

OS- Windows 10

Version – 15.18362

Layer Name – WindowsIntel32e

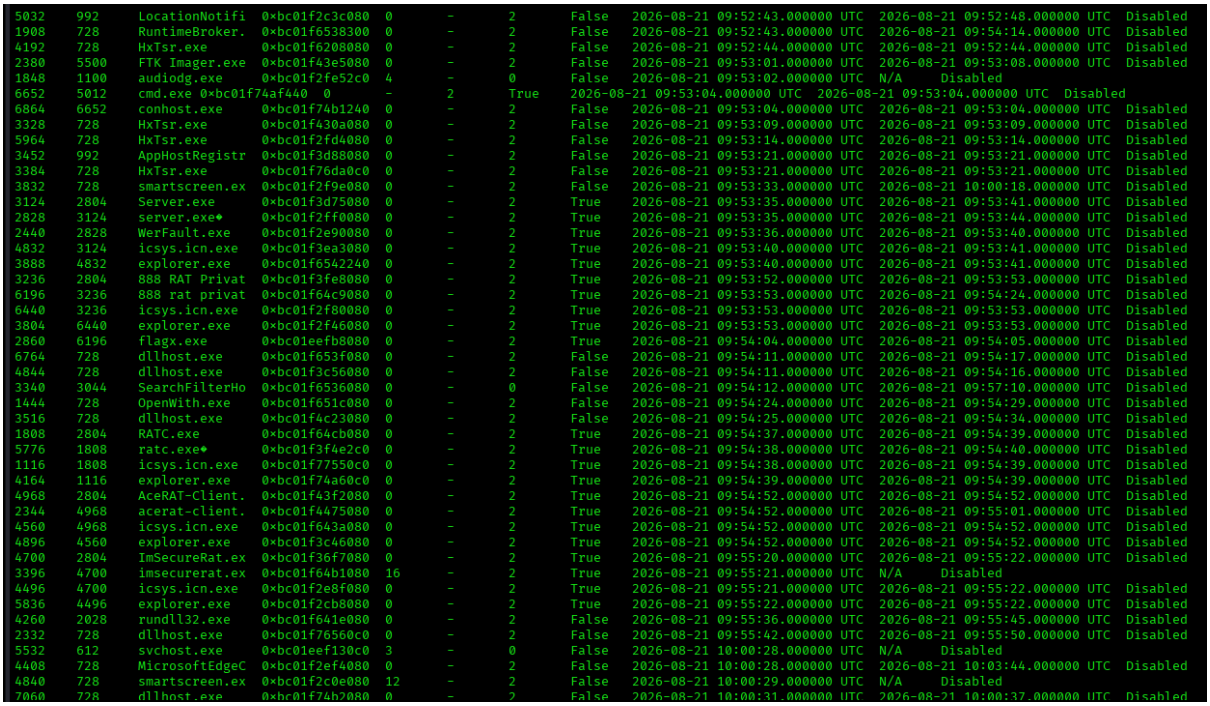
Processors – 2

It did not have a minor Operating System installed in

Figure 6.4

```
┌──(mrvoidhalo㉿kali)-[~/volatility3]
└─$ python3 vol.py -f ~/Desktop/memory/win10_ram.raw windows.pslist
Volatility 3 Framework 2.28.2
Progress:  100.00               PDB scanning finished
PID     PPID    ImageFileName   Offset(V)       Threads Handles SessionId       Wow64   CreateTime      ExitTime        File output

4       0       System  0×bc01eec88380  116     -       N/A     False   2026-08-21 18:28:53.000000 UTC  N/A     Disabled
88      4       Registry        0×bc01eeced080  4       -       N/A     False   2026-08-21 18:28:48.000000 UTC  N/A     Disabled
292     4       smss.exe        0×bc01f0d7e040  2       -       N/A     False   2026-08-21 18:28:53.000000 UTC  N/A     Disabled
400     388     csrss.exe       0×bc01f0d0b080  10      -       0       False   2026-08-21 18:28:54.000000 UTC  N/A     Disabled
476     388     wininit.exe     0×bc01f0f89080  1       -       0       False   2026-08-21 18:28:55.000000 UTC  N/A     Disabled
612     476     services.exe    0×bc01f22f1080  6       -       0       False   2026-08-21 18:28:55.000000 UTC  N/A     Disabled
628     476     lsass.exe       0×bc01f0f36280  10      -       0       False   2026-08-21 18:28:55.000000 UTC  N/A     Disabled
728     612     svchost.exe     0×bc01f31aa240  22      -       0       False   2026-08-21 18:28:55.000000 UTC  N/A     Disabled
756     476     fontdrvhost.ex  0×bc01f31b5140  5       -       0       False   2026-08-21 18:28:55.000000 UTC  N/A     Disabled
848     612     svchost.exe     0×bc01f31d82c0  9       -       0       False   2026-08-21 18:28:55.000000 UTC  N/A     Disabled
912     612     sppsvc.exe      0×bc01f327c080  0       -       0       False   2026-08-21 18:28:55.000000 UTC  2026-08-21 09:29:30.000000 UTC  Disabled
992     612     svchost.exe     0×bc01f32e8240  43      -       0       False   2026-08-21 18:28:55.000000 UTC  N/A     Disabled
324     612     svchost.exe     0×bc01f33272c0  14      -       0       False   2026-08-21 18:28:56.000000 UTC  N/A     Disabled
472     612     svchost.exe     0×bc01f33452c0  19      -       0       False   2026-08-21 18:28:56.000000 UTC  N/A     Disabled
1028    64      msoobe.exe      0×bc01eed8b080  0       -       1       False   2026-08-21 18:28:59.000000 UTC  2026-08-21 09:29:11.000000 UTC  Disabled
1052    612     svchost.exe     0×bc01f0ec1080  16      -       0       False   2026-08-21 18:28:59.000000 UTC  N/A     Disabled
1100    612     svchost.exe     0×bc01eed8d080  9       -       0       False   2026-08-21 18:29:00.000000 UTC  N/A     Disabled
1180    612     svchost.exe     0×bc01eed6b080  17      -       0       False   2026-08-21 18:29:01.000000 UTC  N/A     Disabled
1228    612     svchost.exe     0×bc01f0de92c0  19      -       0       False   2026-08-21 18:29:01.000000 UTC  N/A     Disabled
1816    4       MemCompression  0×bc01f365e040  242     -       N/A     False   2026-08-21 18:29:02.000000 UTC  N/A     Disabled
1860    612     svchost.exe     0×bc01f3678080  3       -       0       False   2026-08-21 18:29:02.000000 UTC  N/A     Disabled
1868    612     svchost.exe     0×bc01f367a080  8       -       0       False   2026-08-21 18:29:02.000000 UTC  N/A     Disabled
1968    612     spoolsv.exe     0×bc01f36d2200  8       -       0       False   2026-08-21 18:29:02.000000 UTC  N/A     Disabled
2028    612     svchost.exe     0×bc01f36f02c0  12      -       0       False   2026-08-21 18:29:02.000000 UTC  N/A     Disabled
2124    612     svchost.exe     0×bc01f3705240  13      -       0       False   2026-08-21 18:29:02.000000 UTC  N/A     Disabled
2160    612     OfficeClickToR  0×bc01f3750340  16      -       0       False   2026-08-21 18:29:03.000000 UTC  N/A     Disabled
2232    612     MsMpEng.exe     0×bc01f3777280  8       -       0       False   2026-08-21 18:29:03.000000 UTC  N/A     Disabled
2944    612     svchost.exe     0×bc01f3c5c240  7       -       0       False   2026-08-21 18:29:05.000000 UTC  N/A     Disabled
3044    612     SearchIndexer.  0×bc01f35a50c0  17      -       0       False   2026-08-21 18:29:06.000000 UTC  N/A     Disabled
2140    992     taskhostw.exe   0×bc01f3d7c080  3       -       0       False   2026-08-21 09:29:11.000000 UTC  N/A     Disabled
3940    728     dllhost.exe     0×bc01f3ce4080  4       -       0       False   2026-08-21 09:29:16.000000 UTC  N/A     Disabled
4088    728     WmiPrvSE.exe    0×bc01f3d07080  3       -       0       False   2026-08-21 09:29:21.000000 UTC  N/A     Disabled
2204    728     CloudExperienc  0×bc01f3f9b0c0  4       -       0       False   2026-08-21 09:30:36.000000 UTC  N/A     Disabled
4464    612     svchost.exe     0×bc01f3dc50c0  5       -       0       False   2026-08-21 09:30:50.000000 UTC  N/A     Disabled
500     612     SgrmBroker.exe  0×bc01f3f762c0  3       -       0       False   2026-08-21 09:31:09.000000 UTC  N/A     Disabled
1996    612     svchost.exe     0×bc01f446e2c0  8       -       0       False   2026-08-21 09:31:10.000000 UTC  N/A     Disabled
4848    992     OfficeC2RClien  0×bc01f3e81080  0       -       0       False   2026-08-21 09:35:34.000000 UTC  2026-08-21 09:36:05.000000 UTC  Disabled
948     1052    ctfmon.exe      0×bc01f0871080  8       -       2       False   2026-08-21 09:35:40.000000 UTC  N/A     Disabled
2348    5000    userinit.exe    0×bc01eedb3080  0       -       2       False   2026-08-21 09:35:40.000000 UTC  2026-08-21 09:36:08.000000 UTC  Disabled
2804    2348    explorer.exe    0×bc01f3d86080  69      -       2       False   2026-08-21 09:35:40.000000 UTC  N/A     Disabled
4768    728     dllhost.exe     0×bc01f22724c0  6       -       2       False   2026-08-21 09:35:42.000000 UTC  N/A     Disabled
1000    612     svchost.exe     0×bc01f37ef080  5       -       2       False   2026-08-21 09:35:45.000000 UTC  N/A     Disabled
3280    728     StartMenuExper  0×bc01f3d09080  8       -       2       False   2026-08-21 09:35:50.000000 UTC  N/A     Disabled
3572    728     RuntimeBroker.  0×bc01f3fc4080  6       -       2       False   2026-08-21 09:35:50.000000 UTC  N/A     Disabled
4616    728     SearchUI.exe    0×bc01f4163080  37      -       2       False   2026-08-21 09:35:52.000000 UTC  N/A     Disabled
1676    728     RuntimeBroker.  0×bc01f0fdb080  9       -       2       False   2026-08-21 09:35:53.000000 UTC  N/A     Disabled
2208    612     svchost.exe     0×bc01f0fb3080  4       -       0       False   2026-08-21 09:35:59.000000 UTC  N/A     Disabled
5236    728     ApplicationFra  0×bc01f43a4080  8       -       2       False   2026-08-21 09:36:15.000000 UTC  N/A     Disabled
5268    728     MicrosoftEdge.  0×bc01f5ae1240  0       -       2       False   2026-08-21 09:36:15.000000 UTC  2026-08-21 10:03:45.000000 UTC  Disabled
5352    728     browser_broker  0×bc01f3eb34c0  0       -       2       False   2026-08-21 09:36:15.000000 UTC  2026-08-21 10:03:45.000000 UTC  Disabled
5436    612     svchost.exe     0×bc01f3cf2080  2       -       0       False   2026-08-21 09:36:15.000000 UTC  N/A     Disabled
5540    728     RuntimeBroker.  0×bc01f3de8080  6       -       2       False   2026-08-21 09:36:15.000000 UTC  N/A     Disabled
5632    5540    MicrosoftEdgeS  0×bc01f0f86080  0       -       2       False   2026-08-21 09:36:16.000000 UTC  2026-08-21 10:03:45.000000 UTC  Disabled
5660    728     MicrosoftEdgeC  0×bc01f5a430c0  0       -       2       False   2026-08-21 09:36:16.000000 UTC  2026-08-21 10:01:13.000000 UTC  Disabled
6008    728     dllhost.exe     0×bc01f3e9e080  2       -       2       False   2026-08-21 09:36:55.000000 UTC  N/A     Disabled
496     992     taskhostw.exe   0×bc01f0f8c080  6       -       2       False   2026-08-21 09:36:55.000000 UTC  N/A     Disabled
3388    728     Microsoft.Phot  0×bc01f406f080  12      -       2       False   2026-08-21 09:36:56.000000 UTC  N/A     Disabled
64      728     ShellExperienc  0×bc01f2be7240  14      -       2       False   2026-08-21 09:36:57.000000 UTC  N/A     Disabled
3292    728     RuntimeBroker.  0×bc01f2edd2c0  2       -       2       False   2026-08-21 09:36:58.000000 UTC  N/A     Disabled
5864    992     OfficeC2RClien  0×bc01f2fe1080  0       -       0       False   2026-08-21 09:37:05.000000 UTC  2026-08-21 09:37:25.000000 UTC  Disabled
3164    992     OfficeC2RClien  0×bc01f43eb340  0       -       0       False   2026-08-21 09:37:05.000000 UTC  2026-08-21 09:37:19.000000 UTC  Disabled
3228    2804    SecurityHealth  0×bc01f3fa63c0  1       -       2       False   2026-08-21 09:37:06.000000 UTC  N/A     Disabled
3760    612     SecurityHealth  0×bc01f2fa2380  10      -       0       False   2026-08-21 09:37:06.000000 UTC  N/A     Disabled
6152    728     RuntimeBroker.  0×bc01f3f9f080  1       -       2       False   2026-08-21 09:37:06.000000 UTC  N/A     Disabled
6792    728     RuntimeBroker.  0×bc01f62ed080  5       -       2       False   2026-08-21 09:37:08.000000 UTC  N/A     Disabled
7048    728     MicrosoftEdgeC  0×bc01f62ee080  0       -       2       False   2026-08-21 09:37:10.000000 UTC  2026-08-21 10:03:45.000000 UTC  Disabled
5044    2644    OneDriveSetup.  0×bc01f415a4c0  0       -       2       True    2026-08-21 09:38:56.000000 UTC  2026-08-21 09:39:09.000000 UTC  Disabled
3456    5044    OneDriveSetup.  0×bc01f6441080  0       -       2       True    2026-08-21 09:39:03.000000 UTC  2026-08-21 09:39:09.000000 UTC  Disabled
4432    3456    OneDrive.exe    0×bc01f3868080  25      -       2       True    2026-08-21 09:39:08.000000 UTC  N/A     Disabled
6024    992     OfficeC2RClien  0×bc01f650e0c0  0       -       0       False   2026-08-21 09:40:40.000000 UTC  2026-08-21 09:40:55.000000 UTC  Disabled
3232    728     SystemSettings  0×bc01f2feb4c0  16      -       2       False   2026-08-21 09:41:50.000000 UTC  N/A     Disabled
6848    612     svchost.exe     0×bc01f3cc5080  3       -       0       False   2026-08-21 09:41:52.000000 UTC  N/A     Disabled
4912    728     usocoreworker.  0×bc01f380e080  6       -       0       False   2026-08-21 09:42:03.000000 UTC  N/A     Disabled
1608    728     smartscreen.ex  0×bc01f35c7080  0       -       2       False   2026-08-21 09:42:07.000000 UTC  2026-08-21 09:52:45.000000 UTC  Disabled
1832    728     SecurityHealth  0×bc01f2cbb080  1       -       2       False   2026-08-21 09:42:24.000000 UTC  N/A     Disabled
1692    3044    SearchFilterHo  0×bc01f62e3080  0       -       0       False   2026-08-21 09:46:59.000000 UTC  2026-08-21 09:49:00.000000 UTC  Disabled
3592    2804    Exterro_FTK_Im  0×bc01f439e080  0       -       2       True    2026-08-21 09:47:45.000000 UTC  2026-08-21 09:47:47.000000 UTC  Disabled
568     3592    exterro_ftk_im  0×bc01f2a2a080  0       -       2       True    2026-08-21 09:47:46.000000 UTC  2026-08-21 09:53:04.000000 UTC  Disabled
5012    568     exterro_ftk_im  0×bc01f2fe6080  0       -       2       True    2026-08-21 09:47:46.000000 UTC  2026-08-21 09:53:04.000000 UTC  Disabled
2644    3592    icsys.icn.exe   0×bc01f3f51080  0       -       2       True    2026-08-21 09:47:46.000000 UTC  2026-08-21 09:47:47.000000 UTC  Disabled
1724    2644    explorer.exe    0×bc01f2f50080  2       -       2       True    2026-08-21 09:47:47.000000 UTC  N/A     Disabled
5316    1724    spoolsv.exe     0×bc01f6437080  0       -       2       True    2026-08-21 09:47:47.000000 UTC  2026-08-21 09:47:47.000000 UTC  Disabled
```



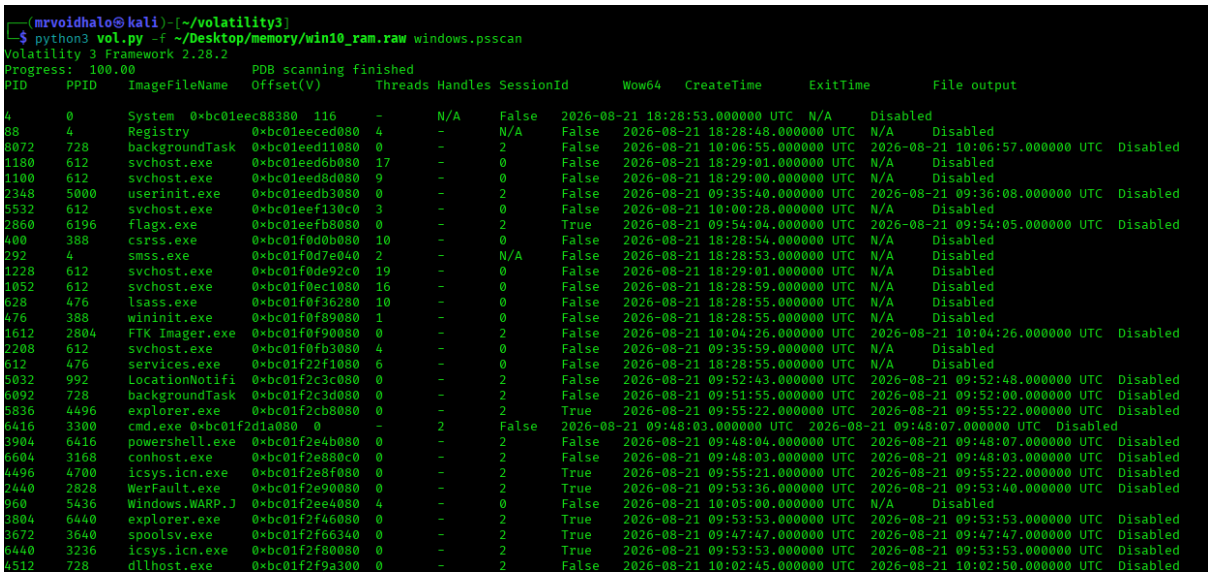
5032	992	LocationNotifi	0×bc01f2c3c080	0	-	2	False	2026-08-21 09:52:43.000000 UTC	2026-08-21 09:52:48.000000 UTC	Disabled
1908	728	RuntimeBroker.	0×bc01f6538300	0	-	2	False	2026-08-21 09:52:43.000000 UTC	2026-08-21 09:54:14.000000 UTC	Disabled
4192	728	HxTsr.exe	0×bc01f6208080	0	-	2	False	2026-08-21 09:52:44.000000 UTC	2026-08-21 09:52:44.000000 UTC	Disabled
2380	5500	FTK Imager.exe	0×bc01f43e5080	0	-	2	False	2026-08-21 09:53:01.000000 UTC	2026-08-21 09:53:08.000000 UTC	Disabled
1848	1100	audiodg.exe	0×bc01f2fe52c0	4	-	0	False	2026-08-21 09:53:02.000000 UTC	N/A	Disabled
6652	5012	cmd.exe	0×bc01f74af440	0	-	2	True	2026-08-21 09:53:04.000000 UTC	2026-08-21 09:53:04.000000 UTC	Disabled
6864	6652	conhost.exe	0×bc01f74b1240	0	-	2	False	2026-08-21 09:53:04.000000 UTC	2026-08-21 09:53:04.000000 UTC	Disabled
3328	728	HxTsr.exe	0×bc01f430a080	0	-	2	False	2026-08-21 09:53:09.000000 UTC	2026-08-21 09:53:09.000000 UTC	Disabled
5964	728	HxTsr.exe	0×bc01f2fd4080	0	-	2	False	2026-08-21 09:53:14.000000 UTC	2026-08-21 09:53:14.000000 UTC	Disabled
3452	992	AppHostRegistr	0×bc01f3d88080	0	-	2	False	2026-08-21 09:53:21.000000 UTC	2026-08-21 09:53:21.000000 UTC	Disabled
3384	728	HxTsr.exe	0×bc01f76da0c0	0	-	2	False	2026-08-21 09:53:21.000000 UTC	2026-08-21 09:53:21.000000 UTC	Disabled
3832	728	smartscreen.ex	0×bc01f2f9e080	0	-	2	False	2026-08-21 09:53:33.000000 UTC	2026-08-21 10:00:18.000000 UTC	Disabled
3124	2804	Server.exe	0×bc01f3d75080	0	-	2	True	2026-08-21 09:53:35.000000 UTC	2026-08-21 09:53:41.000000 UTC	Disabled
2828	3124	server.exe•	0×bc01f2ff0080	0	-	2	True	2026-08-21 09:53:35.000000 UTC	2026-08-21 09:53:44.000000 UTC	Disabled
2440	2828	WerFault.exe	0×bc01f2e90080	0	-	2	True	2026-08-21 09:53:36.000000 UTC	2026-08-21 09:53:40.000000 UTC	Disabled
4832	3124	icsys.icn.exe	0×bc01f3ea3080	0	-	2	True	2026-08-21 09:53:40.000000 UTC	2026-08-21 09:53:41.000000 UTC	Disabled
3888	4832	explorer.exe	0×bc01f6542240	0	-	2	True	2026-08-21 09:53:40.000000 UTC	2026-08-21 09:53:41.000000 UTC	Disabled
3236	2804	888 RAT Privat	0×bc01f3fe8080	0	-	2	True	2026-08-21 09:53:52.000000 UTC	2026-08-21 09:53:53.000000 UTC	Disabled
6196	3236	888 rat privat	0×bc01f64c9080	0	-	2	True	2026-08-21 09:53:53.000000 UTC	2026-08-21 09:54:24.000000 UTC	Disabled
6440	3236	icsys.icn.exe	0×bc01f2f80080	0	-	2	True	2026-08-21 09:53:53.000000 UTC	2026-08-21 09:53:53.000000 UTC	Disabled
3804	6440	explorer.exe	0×bc01f2f46080	0	-	2	True	2026-08-21 09:53:53.000000 UTC	2026-08-21 09:53:53.000000 UTC	Disabled
2860	6196	flagx.exe	0×bc01eefb8080	0	-	2	True	2026-08-21 09:54:04.000000 UTC	2026-08-21 09:54:05.000000 UTC	Disabled
6764	728	dllhost.exe	0×bc01f653f080	0	-	2	False	2026-08-21 09:54:11.000000 UTC	2026-08-21 09:54:17.000000 UTC	Disabled
4844	728	dllhost.exe	0×bc01f3c56080	0	-	2	False	2026-08-21 09:54:11.000000 UTC	2026-08-21 09:54:16.000000 UTC	Disabled
3340	3044	SearchFilterHo	0×bc01f6536080	0	-	0	False	2026-08-21 09:54:12.000000 UTC	2026-08-21 09:57:10.000000 UTC	Disabled
1444	728	OpenWith.exe	0×bc01f651c080	0	-	2	False	2026-08-21 09:54:24.000000 UTC	2026-08-21 09:54:29.000000 UTC	Disabled
3516	728	dllhost.exe	0×bc01f4c23080	0	-	2	False	2026-08-21 09:54:25.000000 UTC	2026-08-21 09:54:34.000000 UTC	Disabled
1808	2804	RATC.exe	0×bc01f64cb080	0	-	2	True	2026-08-21 09:54:37.000000 UTC	2026-08-21 09:54:39.000000 UTC	Disabled
5776	1808	ratc.exe•	0×bc01f3f4e2c0	0	-	2	True	2026-08-21 09:54:38.000000 UTC	2026-08-21 09:54:40.000000 UTC	Disabled
1116	1808	icsys.icn.exe	0×bc01f77550c0	0	-	2	True	2026-08-21 09:54:38.000000 UTC	2026-08-21 09:54:39.000000 UTC	Disabled
4164	1116	explorer.exe	0×bc01f74a60c0	0	-	2	True	2026-08-21 09:54:39.000000 UTC	2026-08-21 09:54:39.000000 UTC	Disabled
4968	2804	AceRAT-Client.	0×bc01f43f2080	0	-	2	True	2026-08-21 09:54:52.000000 UTC	2026-08-21 09:54:52.000000 UTC	Disabled
2344	4968	acerat-client.	0×bc01f4475080	0	-	2	True	2026-08-21 09:54:52.000000 UTC	2026-08-21 09:55:01.000000 UTC	Disabled
4560	4968	icsys.icn.exe	0×bc01f643a080	0	-	2	True	2026-08-21 09:54:52.000000 UTC	2026-08-21 09:54:52.000000 UTC	Disabled
4896	4560	explorer.exe	0×bc01f3c46080	0	-	2	True	2026-08-21 09:54:52.000000 UTC	2026-08-21 09:54:52.000000 UTC	Disabled
4700	2804	ImSecureRat.ex	0×bc01f36f7080	0	-	2	True	2026-08-21 09:55:20.000000 UTC	2026-08-21 09:55:22.000000 UTC	Disabled
3396	4700	imsecurerat.ex	0×bc01f64b1080	16	-	2	True	2026-08-21 09:55:21.000000 UTC	N/A	Disabled
4496	4700	icsys.icn.exe	0×bc01f2e8f080	0	-	2	True	2026-08-21 09:55:21.000000 UTC	2026-08-21 09:55:22.000000 UTC	Disabled
5836	4496	explorer.exe	0×bc01f2cb8080	0	-	2	True	2026-08-21 09:55:22.000000 UTC	2026-08-21 09:55:22.000000 UTC	Disabled
4260	2028	rundll32.exe	0×bc01f641e080	0	-	2	False	2026-08-21 09:55:36.000000 UTC	2026-08-21 09:55:45.000000 UTC	Disabled
2332	728	dllhost.exe	0×bc01f76560c0	0	-	2	False	2026-08-21 09:55:42.000000 UTC	2026-08-21 09:55:50.000000 UTC	Disabled
5532	612	svchost.exe	0×bc01eef130c0	3	-	0	False	2026-08-21 10:00:28.000000 UTC	N/A	Disabled
4408	728	MicrosoftEdgeC	0×bc01f2ef4080	0	-	2	False	2026-08-21 10:00:28.000000 UTC	2026-08-21 10:03:44.000000 UTC	Disabled
4840	728	smartscreen.ex	0×bc01f2c0e080	12	-	2	False	2026-08-21 10:00:29.000000 UTC	N/A	Disabled
7060	728	dllhost.exe	0×bc01f74b2080	0	-	2	False	2026-08-21 10:00:31.000000 UTC	2026-08-21 10:00:37.000000 UTC	Disabled

The windows.pslist command during the analysis there was an unusual parent-child relationships, unexpected active sessions and anomalous timestamps.

Suspicious Parent-Child Process (888 RAT Private-2804, ratc.exe-1808)

The anomaly which this are Remote Access Trojan (RAT) and these ones are the servers which the suspect was trying to inject a malware to the two Victims above. The RAT gives the attacker a full access to the machine whenever the victims machine has access to the internet, in the sessionId 2 session were made now trying to trace the session which are linked to it

Figure 6.5



```
┌──(mrvoidhalo㉿kali)-[~/volatility3]
└─$ python3 vol.py -f ~/Desktop/memory/win10_ram.raw windows.psscan
Volatility 3 Framework 2.28.2
Progress:  100.00               PDB scanning finished
```

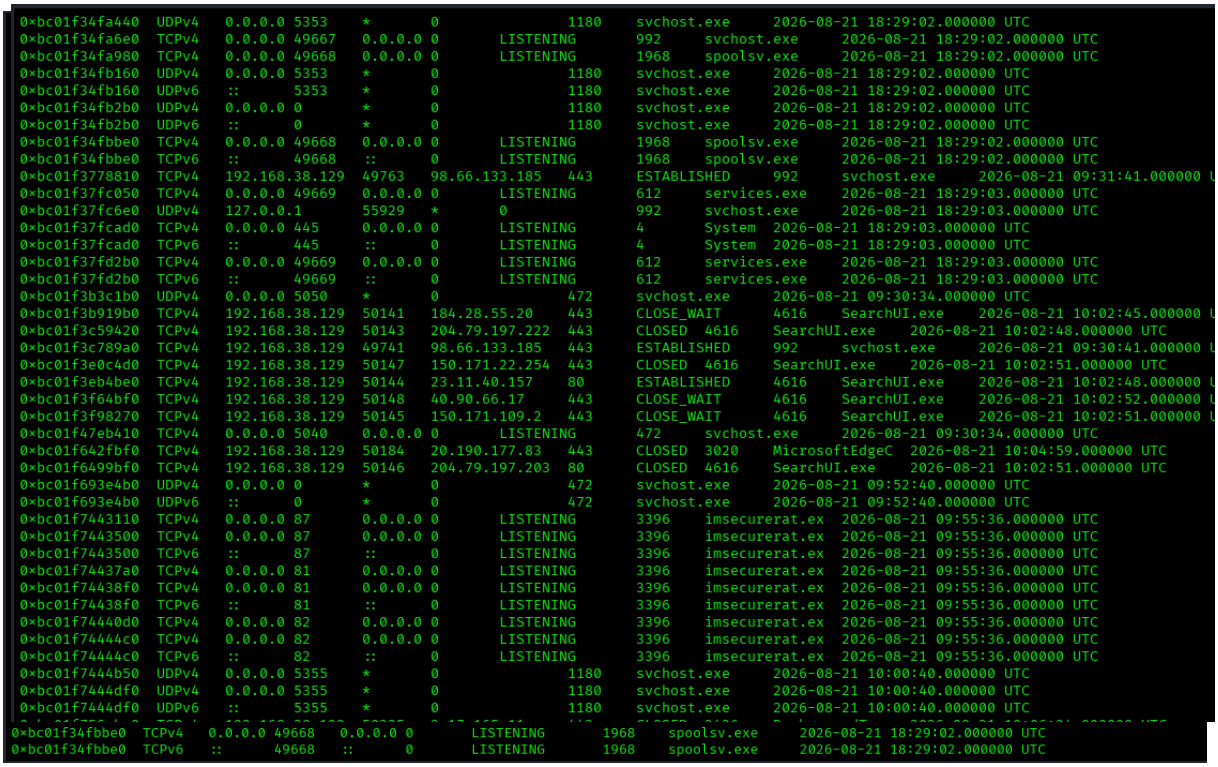
PID	PPID	ImageFileName	Offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime	File output
4	0	System	0×bc01eec88380	116	-	N/A	False	2026-08-21 18:28:53.000000 UTC	N/A	Disabled
88	4	Registry	0×bc01eeced080	4	-	N/A	False	2026-08-21 18:28:48.000000 UTC	N/A	Disabled
8072	728	backgroundTask	0×bc01eed11080	0	-	2	False	2026-08-21 10:06:55.000000 UTC	2026-08-21 10:06:57.000000 UTC	Disabled
1180	612	svchost.exe	0×bc01eed6b080	17	-	0	False	2026-08-21 18:29:01.000000 UTC	N/A	Disabled
1100	612	svchost.exe	0×bc01eed8d080	9	-	0	False	2026-08-21 18:29:00.000000 UTC	N/A	Disabled
2348	5000	userinit.exe	0×bc01eedb3080	0	-	2	False	2026-08-21 09:35:40.000000 UTC	2026-08-21 09:36:08.000000 UTC	Disabled
5532	612	svchost.exe	0×bc01eef130c0	3	-	0	False	2026-08-21 10:00:28.000000 UTC	N/A	Disabled
2860	6196	flagx.exe	0×bc01eefb8080	0	-	2	True	2026-08-21 09:54:04.000000 UTC	2026-08-21 09:54:05.000000 UTC	Disabled
400	388	csrss.exe	0×bc01f0d0b080	10	-	0	False	2026-08-21 18:28:54.000000 UTC	N/A	Disabled
292	4	smss.exe	0×bc01f0d7e040	2	-	N/A	False	2026-08-21 18:28:53.000000 UTC	N/A	Disabled
1228	612	svchost.exe	0×bc01f0de92c0	19	-	0	False	2026-08-21 18:29:01.000000 UTC	N/A	Disabled
1052	612	svchost.exe	0×bc01f0ec1080	16	-	0	False	2026-08-21 18:28:59.000000 UTC	N/A	Disabled
628	476	lsass.exe	0×bc01f0f36280	10	-	0	False	2026-08-21 18:28:55.000000 UTC	N/A	Disabled
476	388	wininit.exe	0×bc01f0f89080	1	-	0	False	2026-08-21 18:28:55.000000 UTC	N/A	Disabled
1612	2804	FTK Imager.exe	0×bc01f0f90080	0	-	2	False	2026-08-21 10:04:26.000000 UTC	2026-08-21 10:04:26.000000 UTC	Disabled
2208	612	svchost.exe	0×bc01f0fb3080	4	-	0	False	2026-08-21 09:35:59.000000 UTC	N/A	Disabled
612	476	services.exe	0×bc01f22f1080	6	-	0	False	2026-08-21 18:28:55.000000 UTC	N/A	Disabled
5032	992	LocationNotifi	0×bc01f2c3c080	0	-	2	False	2026-08-21 09:52:43.000000 UTC	2026-08-21 09:52:48.000000 UTC	Disabled
6092	728	backgroundTask	0×bc01f2c3d080	0	-	2	False	2026-08-21 09:51:55.000000 UTC	2026-08-21 09:52:00.000000 UTC	Disabled
5836	4496	explorer.exe	0×bc01f2cb8080	0	-	2	True	2026-08-21 09:55:22.000000 UTC	2026-08-21 09:55:22.000000 UTC	Disabled
6416	3300	cmd.exe	0×bc01f2d1a080	0	-	2	False	2026-08-21 09:48:03.000000 UTC	2026-08-21 09:48:07.000000 UTC	Disabled
3904	6416	powershell.exe	0×bc01f2e4b080	0	-	2	False	2026-08-21 09:48:04.000000 UTC	2026-08-21 09:48:07.000000 UTC	Disabled
6604	3168	conhost.exe	0×bc01f2e880c0	0	-	2	False	2026-08-21 09:48:03.000000 UTC	2026-08-21 09:48:03.000000 UTC	Disabled
4496	4700	icsys.icn.exe	0×bc01f2e8f080	0	-	2	True	2026-08-21 09:55:21.000000 UTC	2026-08-21 09:55:22.000000 UTC	Disabled
2440	2828	WerFault.exe	0×bc01f2e90080	0	-	2	True	2026-08-21 09:53:36.000000 UTC	2026-08-21 09:53:40.000000 UTC	Disabled
960	5436	Windows.WARP.J	0×bc01f2ee4080	4	-	0	False	2026-08-21 10:05:00.000000 UTC	N/A	Disabled
3804	6440	explorer.exe	0×bc01f2f46080	0	-	2	True	2026-08-21 09:53:53.000000 UTC	2026-08-21 09:53:53.000000 UTC	Disabled
3672	3640	spoolsv.exe	0×bc01f2f66340	0	-	2	True	2026-08-21 09:47:47.000000 UTC	2026-08-21 09:47:47.000000 UTC	Disabled
6440	3236	icsys.icn.exe	0×bc01f2f80080	0	-	2	True	2026-08-21 09:53:53.000000 UTC	2026-08-21 09:53:53.000000 UTC	Disabled
4512	728	dllhost.exe	0×bc01f2f9a300	0	-	2	False	2026-08-21 10:02:45.000000 UTC	2026-08-21 10:02:50.000000 UTC	Disabled

```
1848    1100    audiodg.exe     0×bc01f2fe52c0  4       -       0       False   2026-08-21 09:53:02.000000 UTC  N/A     Disabled
2828    3124    server.exe◆     0×bc01f2ff0080  0       -       2       True    2026-08-21 09:53:35.000000 UTC  2026-08-21 09:53:44.000000 UTC  Disabled
728     612     svchost.exe     0×bc01f31aa240  22      -       0       False   2026-08-21 18:28:55.000000 UTC  N/A     Disabled
756     476     fontdrvhost.ex  0×bc01f31b5140  5       -       0       False   2026-08-21 18:28:55.000000 UTC  N/A     Disabled
848     612     svchost.exe     0×bc01f31d82c0  9       -       0       False   2026-08-21 18:28:55.000000 UTC  N/A     Disabled
992     612     svchost.exe     0×bc01f32e8240  43      -       0       False   2026-08-21 18:28:55.000000 UTC  N/A     Disabled
324     612     svchost.exe     0×bc01f33272c0  14      -       0       False   2026-08-21 18:28:56.000000 UTC  N/A     Disabled
472     612     svchost.exe     0×bc01f33452c0  19      -       0       False   2026-08-21 18:28:56.000000 UTC  N/A     Disabled
7696    728     dllhost.exe     0×bc01f3367080  0       -       2       False   2026-08-21 10:05:09.000000 UTC  2026-08-21 10:05:15.000000 UTC  Disabled
3044    612     SearchIndexer.  0×bc01f35a50c0  17      -       0       False   2026-08-21 18:29:06.000000 UTC  N/A     Disabled
1816    4       MemCompression  0×bc01f365e040  242     -       N/A     False   2026-08-21 18:29:02.000000 UTC  N/A     Disabled
1860    612     svchost.exe     0×bc01f3678080  3       -       0       False   2026-08-21 18:29:02.000000 UTC  N/A     Disabled
1868    612     svchost.exe     0×bc01f367a080  8       -       0       False   2026-08-21 18:29:02.000000 UTC  N/A     Disabled
7500    728     BackgroundTran  0×bc01f36a6080  0       -       2       False   2026-08-21 10:06:23.000000 UTC  2026-08-21 10:06:23.000000 UTC  Disabled
1968    612     spoolsv.exe     0×bc01f36d2200  8       -       0       False   2026-08-21 18:29:02.000000 UTC  N/A     Disabled
2028    612     svchost.exe     0×bc01f36f02c0  12      -       0       False   2026-08-21 18:29:02.000000 UTC  N/A     Disabled
4700    2804    ImSecureRat.ex  0×bc01f36f7080  0       -       2       True    2026-08-21 09:55:20.000000 UTC  2026-08-21 09:55:22.000000 UTC  Disabled
2124    612     svchost.exe     0×bc01f3705240  13      -       0       False   2026-08-21 18:29:02.000000 UTC  N/A     Disabled
2160    612     OfficeClickToR  0×bc01f3750340  16      -       0       False   2026-08-21 18:29:03.000000 UTC  N/A     Disabled
2232    612     MsMpEng.exe     0×bc01f3777280  8       -       0       False   2026-08-21 18:29:03.000000 UTC  N/A     Disabled
4912    728     usocoreworker.  0×bc01f380e080  6       -       0       False   2026-08-21 09:42:03.000000 UTC  N/A     Disabled
6192    728     MicrosoftEdgeC  0×bc01f3be1080  0       -       2       False   2026-08-21 10:01:06.000000 UTC  2026-08-21 10:01:16.000000 UTC  Disabled
4896    4560    explorer.exe    0×bc01f3c46080  0       -       2       True    2026-08-21 09:54:52.000000 UTC  2026-08-21 09:54:52.000000 UTC  Disabled
4844    728     dllhost.exe     0×bc01f3c56080  0       -       2       False   2026-08-21 09:54:11.000000 UTC  2026-08-21 09:54:16.000000 UTC  Disabled
2944    612     svchost.exe     0×bc01f3c5c240  7       -       0       False   2026-08-21 18:29:05.000000 UTC  N/A     Disabled
3812    4244    csrss.exe       0×bc01f3cb9080  11      -       2       False   2026-08-21 09:35:39.000000 UTC  N/A     Disabled
6848    612     svchost.exe     0×bc01f3cc5080  3       -       0       False   2026-08-21 09:41:52.000000 UTC  N/A     Disabled
3940    728     dllhost.exe     0×bc01f3ce4080  4       -       0       False   2026-08-21 09:29:16.000000 UTC  N/A     Disabled
5436    612     svchost.exe     0×bc01f3cf2080  2       -       0       False   2026-08-21 09:36:15.000000 UTC  N/A     Disabled
4088    728     WmiPrvSE.exe    0×bc01f3d07080  3       -       0       False   2026-08-21 09:29:21.000000 UTC  N/A     Disabled
3124    2804    Server.exe      0×bc01f3d75080  0       -       2       True    2026-08-21 09:53:35.000000 UTC  2026-08-21 09:53:41.000000 UTC  Disabled
2140    992     taskhostw.exe   0×bc01f3d7c080  3       -       0       False   2026-08-21 09:29:11.000000 UTC  N/A     Disabled
3452    992     AppHostRegistr  0×bc01f3d88080  0       -       2       False   2026-08-21 09:53:21.000000 UTC  2026-08-21 09:53:21.000000 UTC  Disabled
7016    728     backgroundTask  0×bc01f3db5080  0       -       2       False   2026-08-21 09:50:40.000000 UTC  2026-08-21 09:50:41.000000 UTC  Disabled
952     728     backgroundTask  0×bc01f3dbf080  0       -       2       False   2026-08-21 09:50:43.000000 UTC  2026-08-21 09:50:43.000000 UTC  Disabled
4464    612     svchost.exe     0×bc01f3dc50c0  5       -       0       False   2026-08-21 09:30:50.000000 UTC  N/A     Disabled
3220    5000    dwm.exe 0×bc01f3e14080  14      -       2       False   2026-08-21 09:35:39.000000 UTC  N/A     Disabled
4848    992     OfficeC2RClien  0×bc01f3e81080  0       -       0       False   2026-08-21 09:35:34.000000 UTC  2026-08-21 09:36:05.000000 UTC  Disabled
4832    3124    icsys.icn.exe   0×bc01f3ea3080  0       -       2       True    2026-08-21 09:53:40.000000 UTC  2026-08-21 09:53:41.000000 UTC  Disabled
5776    1808    ratc.exe◆       0×bc01f3f4e2c0  0       -       2       True    2026-08-21 09:54:38.000000 UTC  2026-08-21 09:54:40.000000 UTC  Disabled
2644    3592    icsys.icn.exe   0×bc01f3f51080  0       -       2       True    2026-08-21 09:47:46.000000 UTC  2026-08-21 09:47:47.000000 UTC  Disabled
500     612     SgrmBroker.exe  0×bc01f3f762c0  3       -       0       False   2026-08-21 09:31:09.000000 UTC  N/A     Disabled
2204    728     CloudExperienc  0×bc01f3f9b0c0  4       -       0       False   2026-08-21 09:30:36.000000 UTC  N/A     Disabled
2652    5000    fontdrvhost.ex  0×bc01f3fdb4c0  5       -       2       False   2026-08-21 09:35:39.000000 UTC  N/A     Disabled
```

The windows.psscan command it's the same as the pslist but it is slower because it makes pool scans in the physical memory for EPROCESS signature and it exposes rootkits hiding via the Direct Kernal Object Manipulation (DKOM)

The ratc.exe pid 1808 and the server.exe 3124 those are marked with a small kite with a question mark icon, shows the Rat was serving in the suspect host machine I the handleID nothing was but tin Wow 64 it was true so it was active in the suspect machine but there was session of the on but going RAT connection to a target.

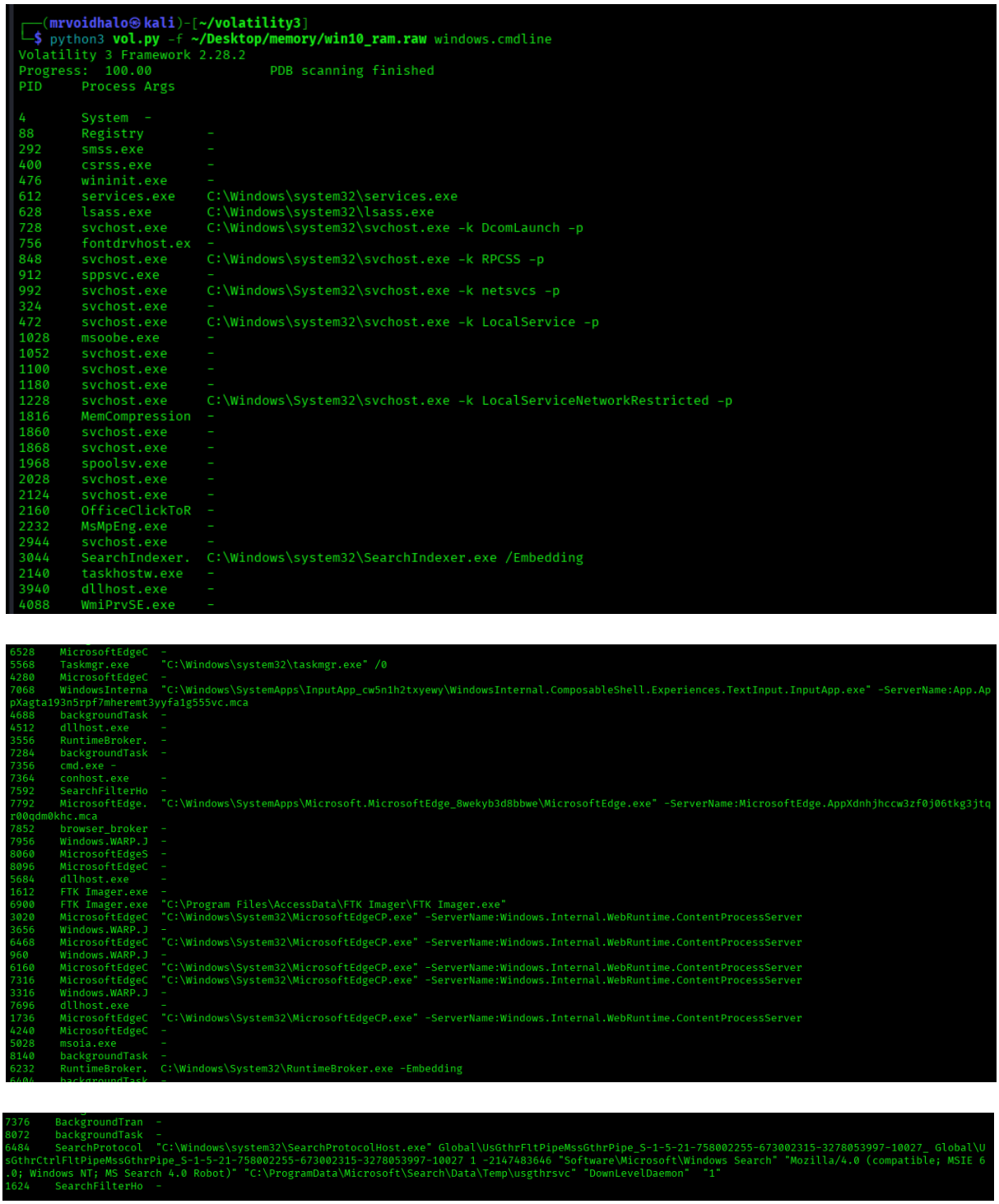
Figure 6.6



```
0×bc01f34fa440  UDPv4   0.0.0.0 5353    *       0               1180    svchost.exe     2026-08-21 18:29:02.000000 UTC
0×bc01f34fa6e0  TCPv4   0.0.0.0 49667   0.0.0.0 0       LISTENING       992     svchost.exe     2026-08-21 18:29:02.000000 UTC
0×bc01f34fa980  TCPv4   0.0.0.0 49668   0.0.0.0 0       LISTENING       1968    spoolsv.exe     2026-08-21 18:29:02.000000 UTC
0×bc01f34fb160  UDPv4   0.0.0.0 5353    *       0               1180    svchost.exe     2026-08-21 18:29:02.000000 UTC
0×bc01f34fb160  UDPv6   ::      5353    *       0               1180    svchost.exe     2026-08-21 18:29:02.000000 UTC
0×bc01f34fb2b0  UDPv4   0.0.0.0 0       *       0               1180    svchost.exe     2026-08-21 18:29:02.000000 UTC
0×bc01f34fb2b0  UDPv6   ::      0       *       0               1180    svchost.exe     2026-08-21 18:29:02.000000 UTC
0×bc01f34fbbe0  TCPv4   0.0.0.0 49668   0.0.0.0 0       LISTENING       1968    spoolsv.exe     2026-08-21 18:29:02.000000 UTC
0×bc01f34fbbe0  TCPv6   ::      49668   ::      0       LISTENING       1968    spoolsv.exe     2026-08-21 18:29:02.000000 UTC
0×bc01f3778810  TCPv4   192.168.38.129  49763   98.66.133.185   443     ESTABLISHED     992     svchost.exe     2026-08-21 09:31:41.000000 U
0×bc01f37fc050  TCPv4   0.0.0.0 49669   0.0.0.0 0       LISTENING       612     services.exe    2026-08-21 18:29:03.000000 UTC
0×bc01f37fc6e0  UDPv4   127.0.0.1       55929   *       0               992     svchost.exe     2026-08-21 18:29:03.000000 UTC
0×bc01f37fcad0  TCPv4   0.0.0.0 445     0.0.0.0 0       LISTENING       4       System  2026-08-21 18:29:03.000000 UTC
0×bc01f37fcad0  TCPv6   ::      445     ::      0       LISTENING       4       System  2026-08-21 18:29:03.000000 UTC
0×bc01f37fd2b0  TCPv4   0.0.0.0 49669   0.0.0.0 0       LISTENING       612     services.exe    2026-08-21 18:29:03.000000 UTC
0×bc01f37fd2b0  TCPv6   ::      49669   ::      0       LISTENING       612     services.exe    2026-08-21 18:29:03.000000 UTC
0×bc01f3b3c1b0  UDPv4   0.0.0.0 5050    *       0               472     svchost.exe     2026-08-21 09:30:34.000000 UTC
0×bc01f3b919b0  TCPv4   192.168.38.129  50141   184.28.55.20    443     CLOSE_WAIT      4616    SearchUI.exe    2026-08-21 10:02:45.000000 U
0×bc01f3c59420  TCPv4   192.168.38.129  50143   204.79.197.222  443     CLOSED  4616    SearchUI.exe    2026-08-21 10:02:48.000000 UTC
0×bc01f3c789a0  TCPv4   192.168.38.129  49741   98.66.133.185   443     ESTABLISHED     992     svchost.exe     2026-08-21 09:30:41.000000 U
0×bc01f3e0c4d0  TCPv4   192.168.38.129  50147   150.171.22.254  443     CLOSED  4616    SearchUI.exe    2026-08-21 10:02:51.000000 UTC
0×bc01f3eb4be0  TCPv4   192.168.38.129  50144   23.11.40.157    80      ESTABLISHED     4616    SearchUI.exe    2026-08-21 10:02:48.000000 U
0×bc01f3f64bf0  TCPv4   192.168.38.129  50148   40.90.66.17     443     CLOSE_WAIT      4616    SearchUI.exe    2026-08-21 10:02:52.000000 U
0×bc01f3f98270  TCPv4   192.168.38.129  50145   150.171.109.2   443     CLOSE_WAIT      4616    SearchUI.exe    2026-08-21 10:02:51.000000 U
0×bc01f47eb410  TCPv4   0.0.0.0 5040    0.0.0.0 0       LISTENING       472     svchost.exe     2026-08-21 09:30:34.000000 UTC
0×bc01f642fbf0  TCPv4   192.168.38.129  50184   20.190.177.83   443     CLOSED  3020    MicrosoftEdgeC  2026-08-21 10:04:59.000000 UTC
0×bc01f6499bf0  TCPv4   192.168.38.129  50146   204.79.197.203  80      CLOSED  4616    SearchUI.exe    2026-08-21 10:02:51.000000 UTC
0×bc01f693e4b0  UDPv4   0.0.0.0 0       *       0               472     svchost.exe     2026-08-21 09:52:40.000000 UTC
0×bc01f693e4b0  UDPv6   ::      0       *       0               472     svchost.exe     2026-08-21 09:52:40.000000 UTC
0×bc01f7443110  TCPv4   0.0.0.0 87      0.0.0.0 0       LISTENING       3396    imsecurerat.ex  2026-08-21 09:55:36.000000 UTC
0×bc01f7443500  TCPv4   0.0.0.0 87      0.0.0.0 0       LISTENING       3396    imsecurerat.ex  2026-08-21 09:55:36.000000 UTC
0×bc01f7443500  TCPv6   ::      87      ::      0       LISTENING       3396    imsecurerat.ex  2026-08-21 09:55:36.000000 UTC
0×bc01f74437a0  TCPv4   0.0.0.0 81      0.0.0.0 0       LISTENING       3396    imsecurerat.ex  2026-08-21 09:55:36.000000 UTC
0×bc01f74438f0  TCPv4   0.0.0.0 81      0.0.0.0 0       LISTENING       3396    imsecurerat.ex  2026-08-21 09:55:36.000000 UTC
0×bc01f74438f0  TCPv6   ::      81      ::      0       LISTENING       3396    imsecurerat.ex  2026-08-21 09:55:36.000000 UTC
0×bc01f74440d0  TCPv4   0.0.0.0 82      0.0.0.0 0       LISTENING       3396    imsecurerat.ex  2026-08-21 09:55:36.000000 UTC
0×bc01f74444c0  TCPv4   0.0.0.0 82      0.0.0.0 0       LISTENING       3396    imsecurerat.ex  2026-08-21 09:55:36.000000 UTC
0×bc01f74444c0  TCPv6   ::      82      ::      0       LISTENING       3396    imsecurerat.ex  2026-08-21 09:55:36.000000 UTC
0×bc01f7444b50  UDPv4   0.0.0.0 5355    *       0               1180    svchost.exe     2026-08-21 10:00:40.000000 UTC
0×bc01f7444df0  UDPv4   0.0.0.0 5355    *       0               1180    svchost.exe     2026-08-21 10:00:40.000000 UTC
0×bc01f7444df0  UDPv6   ::      5355    *       0               1180    svchost.exe     2026-08-21 10:00:40.000000 UTC
0×bc01f34fbbe0  TCPv4   0.0.0.0 49668   0.0.0.0 0       LISTENING       1968    spoolsv.exe     2026-08-21 18:29:02.000000 UTC
0×bc01f34fbbe0  TCPv6   ::      49668   ::      0       LISTENING       1968    spoolsv.exe     2026-08-21 18:29:02.000000 UTC
```

The windows.netscan scans the memory dump for resident network objects, following the insecurerat.ex which is the RAT it is in LISTENING mode where by it was still waiting for receiving connection and it uses TCP both v4 and v6. What is flagging so much it is how the malware keeps changing it offset, when it was waiting for the connection, it kept changing its offset so it was trying to hide its self when the connection was established and would not be easily discovered since it kept changing it offset. The svchost it was functioning well since it a crucial for most malwares tend to hide them selves as scvhost but most common types on they conceal them selves the names change a bit without even noticing (scvh0st,svchost32.exe,scvhost.exe)

Figure 6.7



```
┌──(mrvoidhalo㉿kali)-[~/volatility3]
└─$ python3 vol.py -f ~/Desktop/memory/win10_ram.raw windows.cmdline
Volatility 3 Framework 2.28.2
Progress:  100.00               PDB scanning finished
PID     Process Args

4       System  -
88      Registry        -
292     smss.exe        -
400     csrss.exe       -
476     wininit.exe     -
612     services.exe    C:\Windows\system32\services.exe
628     lsass.exe       C:\Windows\system32\lsass.exe
728     svchost.exe     C:\Windows\system32\svchost.exe -k DcomLaunch -p
756     fontdrvhost.ex  -
848     svchost.exe     C:\Windows\system32\svchost.exe -k RPCSS -p
912     sppsvc.exe      -
992     svchost.exe     C:\Windows\System32\svchost.exe -k netsvcs -p
324     svchost.exe     -
472     svchost.exe     C:\Windows\system32\svchost.exe -k LocalService -p
1028    msoobe.exe      -
1052    svchost.exe     -
1100    svchost.exe     -
1180    svchost.exe     -
1228    svchost.exe     C:\Windows\System32\svchost.exe -k LocalServiceNetworkRestricted -p
1816    MemCompression  -
1860    svchost.exe     -
1868    svchost.exe     -
1968    spoolsv.exe     -
2028    svchost.exe     -
2124    svchost.exe     -
2160    OfficeClickToR  -
2232    MsMpEng.exe     -
2944    svchost.exe     -
3044    SearchIndexer.  C:\Windows\system32\SearchIndexer.exe /Embedding
2140    taskhostw.exe   -
3940    dllhost.exe     -
4088    WmiPrvSE.exe    -
```

```
6528    MicrosoftEdgeC  -
5568    Taskmgr.exe     "C:\Windows\system32\taskmgr.exe" /0
4280    MicrosoftEdgeC  -
7068    WindowsInterna  "C:\Windows\SystemApps\InputApp_cw5n1h2txyewy\WindowsInternal.ComposableShell.Experiences.TextInput.InputApp.exe" -ServerName:App.Ap
pXagta193n5rpf7mheremt3yyfa1g555vc.mca
4688    backgroundTask  -
4512    dllhost.exe     -
3556    RuntimeBroker.  -
7284    backgroundTask  -
7356    cmd.exe -
7364    conhost.exe     -
7592    SearchFilterHo  -
7792    MicrosoftEdge.  "C:\Windows\SystemApps\Microsoft.MicrosoftEdge_8wekyb3d8bbwe\MicrosoftEdge.exe" -ServerName:MicrosoftEdge.AppXdnhjhccw3zf0j06tkg3jtq
r00qdm0khc.mca
7852    browser_broker  -
7956    Windows.WARP.J  -
8060    MicrosoftEdgeS  -
8096    MicrosoftEdgeC  -
5684    dllhost.exe     -
1612    FTK Imager.exe  -
6900    FTK Imager.exe  "C:\Program Files\AccessData\FTK Imager\FTK Imager.exe"
3020    MicrosoftEdgeC  "C:\Windows\System32\MicrosoftEdgeCP.exe" -ServerName:Windows.Internal.WebRuntime.ContentProcessServer
3656    Windows.WARP.J  -
6468    MicrosoftEdgeC  "C:\Windows\System32\MicrosoftEdgeCP.exe" -ServerName:Windows.Internal.WebRuntime.ContentProcessServer
960     Windows.WARP.J  -
6160    MicrosoftEdgeC  "C:\Windows\System32\MicrosoftEdgeCP.exe" -ServerName:Windows.Internal.WebRuntime.ContentProcessServer
7316    MicrosoftEdgeC  "C:\Windows\System32\MicrosoftEdgeCP.exe" -ServerName:Windows.Internal.WebRuntime.ContentProcessServer
3316    Windows.WARP.J  -
7696    dllhost.exe     -
1736    MicrosoftEdgeC  "C:\Windows\System32\MicrosoftEdgeCP.exe" -ServerName:Windows.Internal.WebRuntime.ContentProcessServer
4240    MicrosoftEdgeC  -
5028    msoia.exe       -
8140    backgroundTask  -
6232    RuntimeBroker.  C:\Windows\System32\RuntimeBroker.exe -Embedding
```

```
7376    BackgroundTran  -
8072    backgroundTask  -
6484    SearchProtocol  "C:\Windows\system32\SearchProtocolHost.exe" Global\UsGthrFltPipeMssGthrPipe_S-1-5-21-758002255-673002315-3278053997-10027_ Global\U
sGthrCtrlFltPipeMssGthrPipe_S-1-5-21-758002255-673002315-3278053997-10027 1 -2147483646 "Software\Microsoft\Windows Search" "Mozilla/4.0 (compatible; MSIE 6
.0; Windows NT; MS Search 4.0 Robot)" "C:\ProgramData\Microsoft\Search\Data\Temp\usgthrsvc" "DownLevelDaemon"  "1"
1624    SearchFilterHo  -
```

The windows.cmdline extracts and display the command line arguments that were used to launch each process running in memory. Still trailing the svchost.exe is still not compromised to see if the malware was not concealing itself via the svchost.exe. All the svchost have correct argument of the commands they started in the normal directories.

Summary Analysis

Registry Summary

What user(s) were active?

Two local accounts exist on this machine: Thando Msibi (RID 1002) and Sixole Mkhonta (RID 1003), created minutes apart (SAM hive, samparse plugin - Figure 26). The LogonUI key (Figure 29) confirms Sixole Mkhonta was the account actively logged on at the time of capture, with a LastWrite time of 2026-08-04 08:03:33Z that lines up with the last USB insertion timestamps recorded in Figure 32.

What programs were executed?

This is still largely unanswered from the registry evidence available. The registry confirms Internet Explorer's shell handler is intact and unmodified (Figure 30), and the SOFTWARE/SYSTEM full-plugin reports show no unusual installed services (Figures 34-35, below). However, the UserAssist plugin - normally the strongest registry-based evidence of which programs a user actually launched - returned "not found" (Figures 36-37), and the RecentDocs plugin likewise returned "not found" (Figure 35). This is not evidence that no programs were run; it is an evidence-collection gap. Both checks were run against the Default profile's NTUSER.DAT, because NTUSER.DAT was never captured for either Thando Msibi or Sixole Mkhonta (Figures 22-23), and the Default profile is not used interactively, so it would never contain UserAssist or RecentDocs entries regardless of what Sixole Mkhonta actually did on the machine. Direct evidence of specific programs executed therefore still requires UserAssist, RecentDocs and Shellbags run against Sixole Mkhonta's own NTUSER.DAT, plus the Amcache hive - see "Additional Evidence Still Needed" below.

Evidence of persistence?

One legitimate autostart entry was found: OneDriveSetup under Software\Microsoft\Windows\CurrentVersion\Run (Figure 19), a standard, expected entry on Windows 10 and not evidence of malicious persistence. No entries were found under the legacy load/run values (Figure 20), the StartUp folder has not been redirected (Figure 21), and the Internet Explorer shell handler shows no command-hijacking (Figure 30). The full-plugin SOFTWARE scan also checked the Active Setup, AllowedEnumeration and AMSIProviders keys - all three are common persistence/defense-evasion locations (AMSI providers in particular can be abused to disable antimalware scanning, MITRE T1546), and all three returned "not found" (Figure 34), meaning nothing is registered at those keys. The SYSTEM hive's "svc" plugin was also reviewed (Figures 34-35): the services visible (DiagTrack, DmEnrollmentSvc, EntAppSvc, ESENT, EventSystem) are all built-in Windows services running from the standard %SystemRoot%\\System32\\svchost.exe path with normal service types - none point to an unusual or non-standard executable path. No scheduled tasks were found either. Based on the Run/RunOnce, shell-handler, Active Setup/AMSI, and Services checks performed, no malicious persistence was identified in the registry evidence reviewed so far - however this conclusion remains provisional: only the services visible in the screenshots above were reviewed (the full svc plugin output covers many more), and scheduled tasks are better confirmed from the file system (C:\\Windows\\System32\\Tasks) than from the registry alone (see below).

Memory Summary

What processes existed in memory?
In addition to the normal Windows process tree, two clearly malicious processes were present: **ratc.exe** (PID 1808) and **server.exe** (PID 3124). Both exhibited anomalous parent-child relationships and were recovered by both pslist and the more thorough psscan pool walk. These binaries constitute a Remote Access Trojan implant capable of providing an attacker full remote control of the host once a connection is established.

What network connections were present?

The RAT was observed in LISTENING mode on TCP (both IPv4 and IPv6), waiting for inbound connections. No established external C2 session was captured at the moment of the dump, but the listening socket demonstrates readiness to accept a connection. Offset-shifting behaviour while listening is consistent with anti-forensic techniques intended to hinder detection.

Are there indicators of code injection?

Yes. The presence of unsigned, non-system binaries with anomalous process ancestry, combined with a stealth-oriented network listener, constitutes strong evidence of a malicious implant. A dedicated malfind examination of injected memory regions is recommended to characterise any process-hollowing or reflective-loading techniques that may have been used to deploy the RAT.

Conclusions & Recommendations

The examination establishes that the subject workstation was compromised by a Remote Access Trojan that was active in memory at the time of acquisition. Two local user accounts were created on the morning of the examination date, one of which (Sixole Mkhonta) was interactively logged on. Concurrent USB device activity was recorded. No registry-based persistence was identified in the artefacts available, although critical user-profile hives were missing.

Additional evidence still required / recommended:

1. Capture and analyse the interactive users' NTUSER.DAT and UsrClass.dat hives (UserAssist, RecentDocs, Shellbags, TypedURLs).
2. Examine Amcache.hve and Prefetch files for programme-execution history.
3. Perform a full windows.malfind scan and extract any injected PE sections for further static analysis.
4. Review C:\Windows\System32\Tasks and the full service configuration for residual persistence.
5. Correlate USB serial numbers and volume GUIDs with any external media still in custody.
6. Obtain network logs (if available) covering the acquisition window for any successful inbound connections to the RAT listener.

Report Authentication

This report was prepared by the examiners named on the cover page. All analytical steps described herein were performed in a controlled laboratory environment using the tools and methods documented above. Integrity of the source evidence was verified by SHA-256 hashing prior to analysis.

Examiner: Thando Sibakele Msibi

Scribe: Banel Sixole Mkhonta

Organisation: Advanced School of IT

Date: 04 August 2026

Classification: CONFIDENTIAL